

Female-germline specific protein Sakura interacts with Otu and is crucial for germline stem cell renewal and differentiation and oogenesis

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eLife Assessment

This **valuable** study reports the first characterization of the CG14545 gene in *Drosophila melanogaster*, which the authors name "Sakura." Acting during germline stem cell fate and differentiation, Sakura is required for both oogenesis and female fertility, although some mechanistic details require further investigation. This **solid** study presents a wide-ranging and well-controlled characterization of Sakura, and accordingly the findings and associated reagents described will be of use to scientists interested in oogenesis and early development.

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Abstract

During oogenesis, self-renewal and differentiation of germline stem cells (GSCs) must be tightly regulated. The *Drosophila* female germline serves as an excellent model for studying these regulatory mechanisms. Here, we report that a previously uncharacterized gene *CG14545*, which we named *sakura*, is essential for oogenesis and female fertility in *Drosophila*. Sakura is predominantly expressed in the ovaries, particularly in the germline cells, including GSCs. *sakura* null mutant female flies display rudimentary ovaries with germline-less and tumorous phenotypes, fail to produce eggs, and are completely sterile. The germline-specific depletion of *sakura* impairs Dpp/BMP signaling, leading to aberrant *bag-of-marbles* (*bam*) expression, resulting in faulty differentiation and loss of GSCs. *sakura* is also necessary for normal levels of piwi-interacting RNAs (piRNAs) levels and for female-specific splicing of *sex-lethal* (*sxl*), a master regulator of sex identity determination. We identified Ovarian Tumor (Otu) as a protein binding partner of Sakura and found that loss of *otu* phenocopies loss of *sakura* in ovaries. Thus, we identify Sakura as a crucial factor for GSC renewal and differentiation and oogenesis, and propose that Sakura and Otu function together in these processes.

Introduction

Oogenesis is a complex process in which germline stem cells (GSCs) develop into mature female gametes or oocytes. This process involves multiple layers of regulations and numerous genes, some of which remain to be discovered and characterized. *Drosophila melanogaster*, a genetically tractable model organism, has long been at the forefront of GSC and oogenesis research (Kirilly and Xie 2007 [↗](#)).

Adult *Drosophila* females possess a pair of ovaries, each composed of 12-16 ovarioles. At the anterior tip of each ovariole is a structure called the germarium, which houses two to three GSCs (Fig 1A [↗](#)). GSCs reside in the most anterior region of the germarium, where they directly contact cap cells and escort cells, forming a niche crucial for GSC self-renewal (Lin and Spradling 1993 [↗](#); de Cuevas and Matunis 2011 [↗](#); Losick et al. 2011 [↗](#)). GSCs undergo asymmetric mitotic division, producing two distinct cells: one GSC and one cystoblast. The cystoblast, destined to differentiate, undergoes four mitotic divisions with incomplete cytokinesis, resulting in interconnected cysts of 2, 4, 8, and 16 cells. Of the 16 cells, one will differentiate into an oocyte, while the remaining 15 develop into polyploid nurse cells, which synthesize proteins and RNAs for deposition into the oocyte.

GSCs must be tightly regulated to maintain their undifferentiated state while dividing and differentiating to produce one GSC and one cystoblast. Failure in GSC self-renewal or inappropriate differentiation leads to stem cell loss, disrupting oocyte production (Lin 1997 [↗](#); Cox et al. 1998 [↗](#)). Conversely, uncontrolled GSC division or defective differentiation results in an overabundance of GSC-like cells (a tumorous phenotype) and fewer differentiated cells, ultimately inhibiting oocyte production (Gateff et al. 1996 [↗](#); Ohlstein et al. 2000 [↗](#)). Thus, a precise balance between GSC self-renewal and cyst differentiation within the germarium is essential for proper oogenesis and is tightly regulated.

Bone Morphogenetic Protein (BMP) signaling plays a critical role in regulating GSC self-renewal and differentiation during *Drosophila* oogenesis. In the stem cell niche, GSCs are physically anchored to cap cells via adherens and gap junctions (Song et al. 2002 [↗](#); Gilboa et al. 2003 [↗](#)). Cap cells secrete BMP ligands, such as Decapentaplegic (Dpp) and Glass bottom boat (Gbb), which are recognized by transducing receptors like Thickvein (Tkv) and Saxophone (Sax) on the GSCs (Fig S1 [↗](#)) (Xie and Spradling 1998 [↗](#); Casanueva and Ferguson 2004 [↗](#)). Upon activation, these receptors phosphorylate a transcription factor Mad. The phosphorylated Mad (pMad) translocates into the nucleus, where it represses the transcription of *bag-of-marbles* (*bam*), a key differentiation factor (Kai and Spradling 2003 [↗](#); Song et al. 2004 [↗](#)).

Bam, a ubiquitin-associated protein, is essential for cystoblast differentiation (McKearin and Spradling 1990 [↗](#); McKearin and Ohlstein 1995 [↗](#)). Dpp/BMP signaling in the niche represses *bam* expression, preventing GSCs from differentiating and enabling self-renew (Song et al. 2004 [↗](#)). However, as the daughter cells (cystoblasts) exit the niche, they derepress *bam* and initiate the differentiation program (Fig S1 [↗](#)). Tight regulation of Dpp/BMP signaling and *bam* expression ensures that only cells within the niche adopt GSC fate, while adjacent daughter cells differentiate into cystoblasts. Loss of *bam* leads to blocked differentiation and the accumulation of GSC-like cells (tumorous phenotype), whereas ectopic *bam* expression forces GSCs differentiation, depleting the stem cell pool (McKearin and Spradling 1990 [↗](#); McKearin and Ohlstein 1995 [↗](#); Ohlstein and McKearin 1997 [↗](#)).

One of the proteins that interacts with Bam is Ovarian Tumor (Otu) (Ji et al. 2017 [↗](#)). Otu forms a deubiquitinase complex with Bam, which deubiquitinates Cyclin A (CycA), stabilizing CycA and promoting GSC differentiation (Ji et al. 2017 [↗](#)). Otu also binds RNAs (Ji et al. 2019 [↗](#)). Otu is crucial

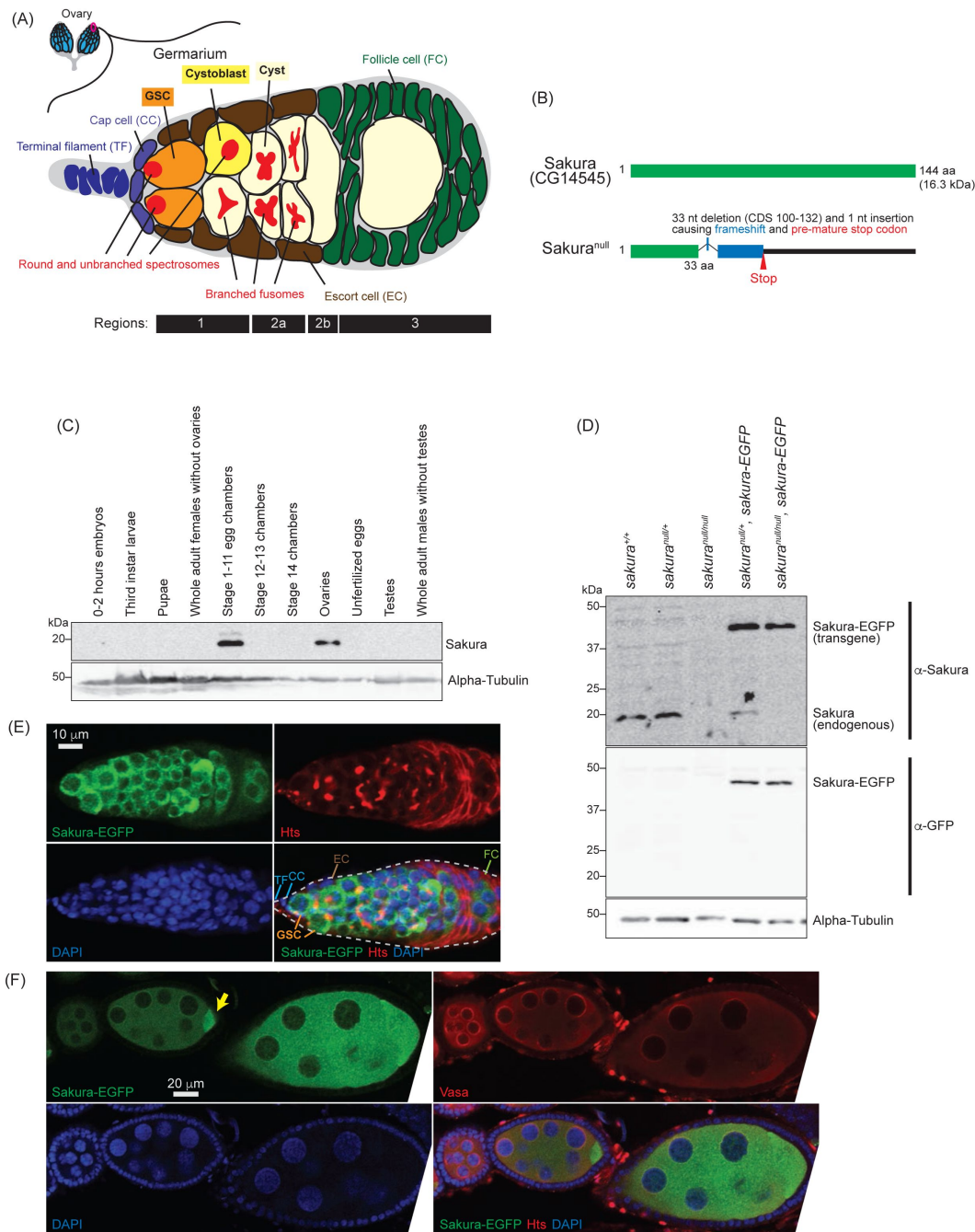


Fig 1.

Sakura expression pattern and mutant allele

(A) Schematic illustration of *Drosophila* ovary and germarium. *Drosophila* female has a pair of ovaries, each consisting of 12-16 ovarioles (cyan). The germarium (outlined with magenta) is located at the anterior tip of the ovarioles and consists of both germ cells and somatic cells. Germ cells include germline stem cells (GSCs), cystoblasts, cysts, and differentiating oocytes. Somatic cells include terminal filament (TF) cells, cap cells (CCs), escort cells (ECs), and follicle cells (FCs). GSCs and cystoblasts have spherical, unbranched spectrosomes, whereas cysts possess branched fusomes. The distinct regions of the germarium—1, 2a, 2b, and 3—are indicated. (B) *Drosophila* Sakura protein (Sakura/CG14545) and its null mutant allele generated in this study (C) Western blot of dissected fly tissues. (D) Western blot of ovary lysates. (E) Confocal images of the germarium from the *sakura*-EGFP transgenic fly. Sakura-EGFP (green), Hts (red), and DAPI (blue). Scale bar: 10 μ m. (F) Confocal images of the egg chambers from the *sakura*-EGFP transgenic fly. Sakura-EGFP (green), Vasa (red), and DAPI (blue). Sakura-EGFP is expressed in nurse cells and enriched in the developing oocyte (yellow arrow). Scale bar: 20 μ m.

for oogenesis and female fertility. Mutations in the *otu* gene cause various ovarian defects, including tumorous growths, germ cell loss, abnormalities in oocyte determination and nurse cell dumping, and defects in the sexual identity determination of germ cells, indicating Otu's importance in multiple stages of oogenesis (Smith and King 1966 [↗](#); Gans et al. 1975 [↗](#); Gollin and King 1981 [↗](#); King and Riley 1982 [↗](#); Storto and King 1988 [↗](#); Pauli et al. 1993 [↗](#); Rodesch et al. 1997 [↗](#); Glenn and Searles 2001 [↗](#)). However, the precise molecular mechanisms by which Otu functions in these processes remain largely unknown.

In this study, we aimed to identify a novel gene required for oogenesis. Through a search for uncharacterized genes expressed exclusively in *Drosophila* ovaries, we discovered *CG14545*, a gene encoding a 114 amino acids (aa) long protein without any known functional domain (**Fig 1B** [↗](#)). We found that *CG14545* is expressed specifically in female germline cells, including GSCs. Our genetic analysis revealed that *CG14545* is required for oogenesis and female fertility and plays an important intrinsic role in regulating GSCs renewal and differentiation. Additionally, *CG14545* is important for the piRNA pathway and for the female-specific mRNA splicing pattern of *sex-lethal* (*sxl*), a master regulator of sex determination. We identified Otu as a binding partner of the *CG14545* protein, suggesting that both proteins may participate in the same molecular pathways to regulate GSC fate and differentiation. We named the *CG14545* gene *sakura*, meaning “cherry blossom” in Japanese, symbolizing birth and renewal.

Results

Sakura is exclusively expressed in the ovaries

Based on high-throughput expression data in Flybase, *sakura* mRNA is highly and exclusively expressed in adult ovaries (**Fig S2** [↗](#)). To investigate Sakura protein expression, we generated a polyclonal anti-Sakura antibody against a recombinant full-length Sakura protein. Using this antibody, we examine Sakura protein expression at different stages of *Drosophila* development and across tissues. We found that Sakura protein is specifically expressed in ovaries, particularly in stage 1-11 egg chambers (**Fig 1C** [↗](#)).

sakura mutant flies

To explore the biological and molecular functions of Sakura *in vivo*, we generated a *sakura* mutant allele (*sakura*^{null}) with a one nucleotide (nt) insertion replacing 33 nts within the Sakura coding region using the CRISPR/Cas9 genome editing system (**Fig 1B** [↗](#)). This mutation causes a frameshift and a premature stop codon, resulting in a 33-amino acid (aa) N-terminal fragment of Sakura followed by a 22-aa segment produced by frameshifted translation (**Fig 1B** [↗](#)). This truncated fragment is unlikely to be functional, and we were unable to detect stable protein expression, as shown below (**Fig 1D** [↗](#)). Thus, we consider this allele a null mutation. *sakura*^{null/null} flies were viable, revealing that Sakura is not essential for survival.

To validate the *sakura*^{null} strain, we performed Western blots using ovary lysates from *sakura* mutant flies using the anti-Sakura antibody. As expected, full-length Sakura protein was detected in the ovaries of wild-type (*sakura*^{+/+}) and heterozygous controls (*sakura*^{null/+}), but not in those of homozygous mutants (*sakura*^{null/null}) (**Fig 1D** [↗](#)). No smaller protein corresponding to the truncated Sakura fragment was detected in either *sakura*^{null/+} or *sakura*^{null/null} ovaries thus, suggesting that the fragment is unstable or not expressed.

Sakura is cytoplasmic and expressed specifically in germ cells

To determine the localization of Sakura protein within the ovaries, we generated transgenic flies carrying a Sakura coding sequence fused with EGFP at the C-terminus (Sakura-EGFP) under the control of the *sakura* promoter. Western blot analysis of ovary lysates showed that Sakura-EGFP is

expressed at levels comparable to endogenous Sakura (Fig 1D).

Oogenesis begins in the germarium, which contains 2-3 GSCs identified by round, unbranched spectrosomes that contact cap cells (Fig 1A) (Kirilly and Xie 2007). In contrast, cysts exhibit branched fusomes. Immunostaining with hu-li tai shao (HTS) antibody marks both spectrosomes and fusomes (Fig 1E), while Vasa is a known germ cell marker (Fig 1F). Confocal imaging revealed that Sakura-EGFP is specifically expressed in the cytoplasm of germ cells, including GSCs, cyst cells, nurse cells, and developing oocytes (Fig 1E and 1F). Sakura-EGFP was not expressed in somatic cells such as terminal filament, cap cells, escort cells, or follicle cells. (Fig 1E). In the egg chamber, Sakura-EGFP was detected in the cytoplasm of nurse cells and was enriched in developing oocytes (Fig 1F).

sakura is essential for female fertility

Given its ovary-specific expression at both mRNA and protein levels (Fig 1C and S2), we hypothesized that Sakura plays a critical role in ovarian function. Fertility assays revealed that *sakura*^{null/null} females laid no eggs when mated with wild-type males, while control females (*sakura*^{+/+} and *sakura*^{null/+}) laid numerous eggs (Fig 2A and 2B). Dissection of *sakura*^{null/null} ovaries revealed that they were rudimentary compared to normal ovaries in controls (Fig 2C). This suggests that the inability of *sakura*^{null/null} female to lay eggs is due to their underdeveloped ovaries. In contrast, *sakura*^{null/null} males were fertile and showed no significant differences compared to controls (*sakura*^{+/+} and *sakura*^{null/+}) (Fig S3).

To confirm that the observed female sterility was caused by the loss of *sakura*, we generated *sakura* rescue flies expressing Sakura-EGFP in the *sakura*^{null/null} background (*sakura*^{null/null}; *sakura-EGFP*). Western blotting confirmed that these flies expressed Sakura-EGFP, but not endogenous Sakura (Fig 1D). The *sakura-EGFP* transgene fully rescued both female fertility and ovary morphology in the *sakura*^{null/null} background (Fig 2). *sakura-EGFP* rescue females showed no significant fertility differences compared to controls (Fig 2A and 2B), and their ovaries were normal in shape and size (Fig 2C). We concluded that Sakura is essential for female fertility and normal ovary morphology but dispensable for male fertility, consistent with the observation that Sakura is predominantly expressed in ovarian germline cells (Fig 1).

sakura^{null/null} ovaries are germless and tumorous

Given the rudimentary appearance of *sakura*^{null/null} ovaries, we questioned whether they contained germ cells. We used flies carrying *vasa-EGFP* knocked-in reporter in the *sakura*^{null/null} background, as Vasa is a known germ cell marker. We observed that some *sakura*^{null/null} ovarioles were devoid of germ cells (“germless”, cyan stars), while others retained germ cells (Fig 3A and 3B). Immunostaining with HTS antibody revealed an excess number of GSC-like cells with round spectrosomes in *sakura*^{null/null} ovarioles containing germ cells (Fig 3C, orange stars), indicative of a “tumorous” phenotype as previously described for mutants of other genes including *bam*, *otu*, and *sxl* (Smith and King 1966; Gans et al. 1975; Gollin and King 1981; King and Riley 1982; McKearin and Spradling 1990; Bopp et al. 1993; Eliazer et al. 2011; Jin et al. 2013; Yang et al. 2019). Additionally, we observed an excess number of cyst cells with branched fusomes that persisted throughout the ovarioles, suggesting abnormal cyst cell differentiation and division. Thus, loss of *sakura* results in both germ cell depletion and overgrowth.

Within the same ovary, some *sakura*^{null/null} ovarioles exhibited the tumorous phenotype, while others were germless, without any spectrosome or fusome staining (Fig 3A-3C, cyan stars). Quantification revealed that 35% of *sakura*^{null/null} ovarioles from 2-to 5-day-old flies were tumorous, while the remaining 65% were germless (n=74) (Fig 3D). In contrast, all ovarioles in control (*sakura*^{+/+} and *sakura*^{null/+}) and *sakura-EGFP* rescue flies were normal with no tumorous or germless phenotypes observed (Fig 3D). Approximately 95% of *sakura*^{null/null} ovarioles

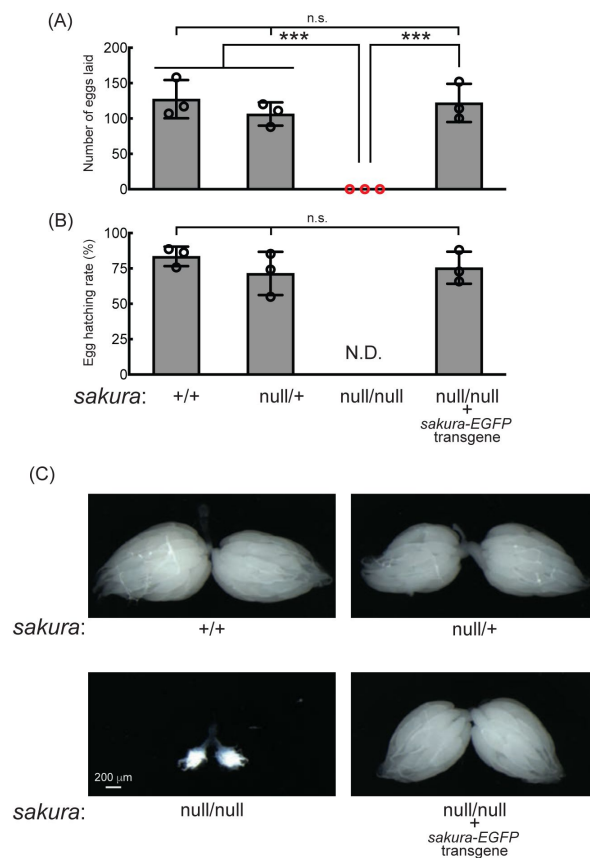


Fig 2.

***sakura* mutant flies are female-sterile and have rudimentary ovaries.**

(A-B) Female fertility assays. (A) The number of eggs laid by test females crossed with OregonR wild-type males and (B) hatching rates of the eggs. Mean $\pm$ SD ($n = 3$). P-value < 0.001 (Student's t-test, unpaired, two-tailed) is indicated by ***. (C) Stereomicroscope images of dissected whole ovaries. Scale bar: 200 μ m.

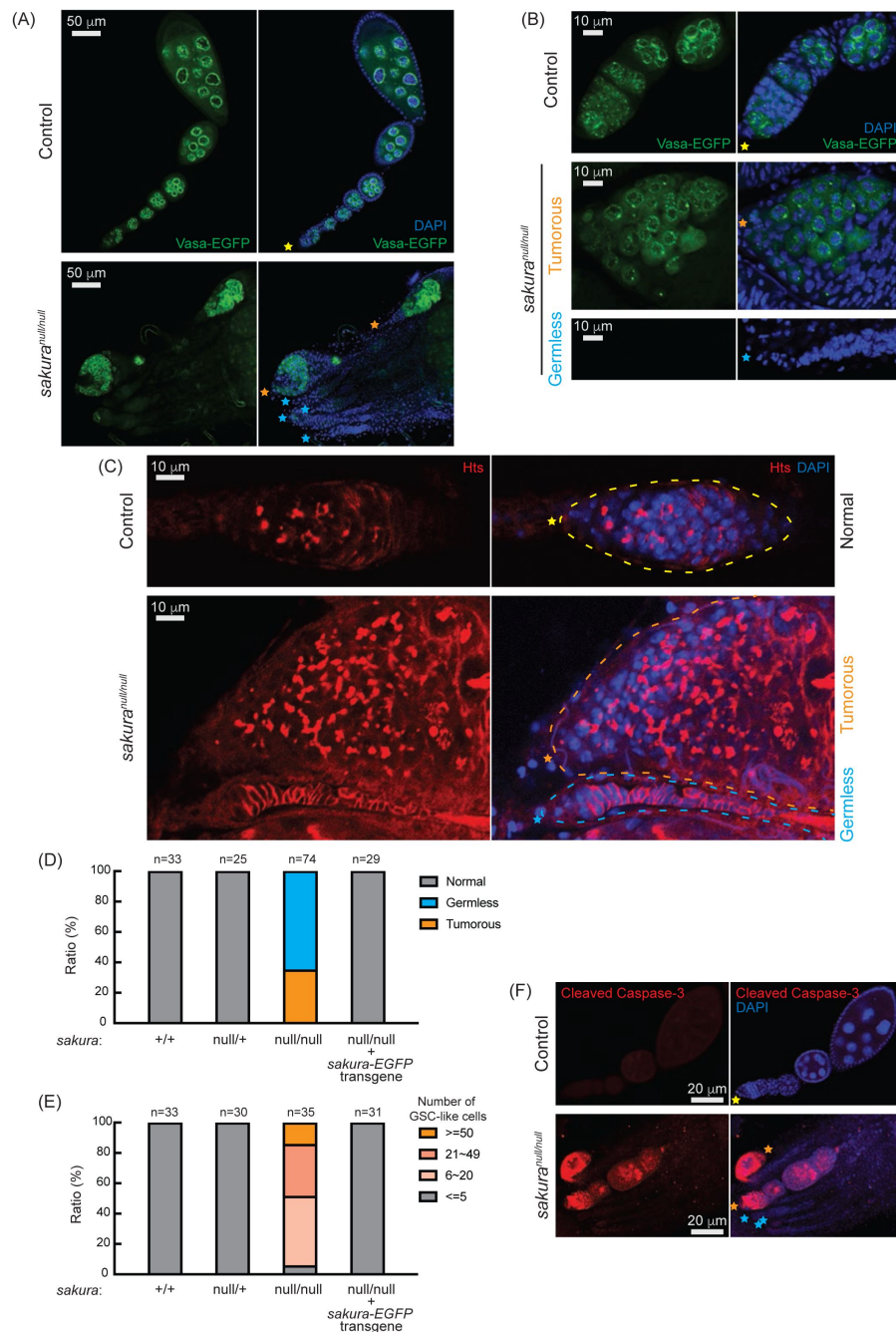


Fig 3.

sakura^{null} ovaries are germless and tumorous

(A, B) Confocal images of the ovaries from control (*sakura*^{null/+}) and *sakura*^{null/null} expressing Vasa-EGFP. Vasa-EGFP (green) and DAPI (blue). Yellow stars indicate the anterior tip of normal ovarioles, while orange and cyan stars indicate the anterior tips of tumorous and germless ovarioles, respectively, in **Figure 1**. Higher-magnification images of the germarium regions are shown in (B). Scale bars: 10 μ m for (A) and 50 μ m for (B). (C) Confocal images of control (*sakura*^{null/+}) and *sakura*^{null/null} ovaries stained with anti-Hts antibody to label spectrosomes and fusomes. Hts (red) and DAPI (blue). Yellow, orange, and cyan dotted lines mark the normal, tumorous, germless and germaria, respectively. Scale bars: 10 μ m. (D) Ratio (%) of normal, germless, and tumorous ovarioles of indicated genotypes (ages 2-5 days; n=33, 25, 74, 29 respectively). (E) Quantification of GSC-like cell number per germarium in the indicated genotypes (ages 2-5 days; n = 33, 30, 35, and 31 respectively). (F) Confocal images of control (*sakura*^{null/+}) and *sakura*^{null/null} ovaries stained with anti-cleaved Caspase-3 antibody. Cleaved caspase-3 (red) and DAPI (blue). Scale bars: 20 μ m.

containing germ cells had an excess of GSC-like cells (>5) (**Fig 3E** and **Fig S4**). The mean number of GSCs or GSC-like cells in 2- to 5-day-old *sakura*^{+/+}, *sakura*^{null/+}, and *sakura-EGFP* rescue was 2.2 ± 0.6 , 2.2 ± 0.6 , and 2.1 ± 0.7 , respectively while that for *sakura*^{null/null} was 26.5 ± 17.5 (p-value <0.05).

We investigated the age-dependent effects on the tumorous and germless phenotypes by examining the ovarioles from 0-1 day old, 7-day old, and 14-day old flies. Ovarioles in control and *sakura-EGFP* rescue flies remained normal throughout this time course (**Fig S5A**). In the very young, 0-1 day old *sakura*^{null/null}, 78% of ovarioles were tumorous and 22% were germless. Overtime, the ratio of germless ovarioles increased while the ratio of tumorous ovarioles decreased. By 14 days old, only 5% of ovarioles were tumorous, with the remaining 95% being germless. The number of GSC-like cells in *sakura*^{null/null} ovarioles was already high in 0-1 day old flies, and it decreased over time (**Fig S5B**). Tumorous ovarioles, but not germless ones, exhibited significantly higher levels of cleaved Caspase-3 staining, indicative of apoptosis, compared to controls (**Fig 3F**). These results suggest that tumorous ovarioles undergo apoptosis, and that these ovarioles eventually become germless.

Together, these results suggest that Sakura is essential for regulating the survival, proliferation, and differentiation of germline cells, including GSCs.

Loss of *sakura* results in reduced germline piRNA

In controls, Vasa-EGFP is enriched in the perinuclear structure known as the nuage within nurse cells (**Fig 3A** and **3B**), which is essential for piwi-interacting RNA (piRNA) biogenesis. In *sakura*^{null/null} tumorous ovarioles, Vasa-EGFP retains its perinuclear localization, indicating that *sakura* is not required for the proper localization of Vasa to the nuage.

piRNAs are produced in germline cells and somatic follicle cells and transcriptionally and post-transcriptionally silence transposon expression through sequence-complementarity (Huang et al. 2017; Yamashiro and Siomi 2018). Disruption of the piRNA pathway can result in oogenesis arrest, germ cell loss, rudimentary ovaries, and sterility. Loss of piRNAs in germ cells leads to the upregulation (desilencing) of transposons, increased DNA damage, and apoptotic cell death³¹. We investigated whether the loss of *sakura* would result in loss of piRNA and transposon upregulation. We performed high-throughput sequencing (small RNA-seq and polyA+ RNA-seq) on ovary RNA samples from *sakura*^{+/+}, *sakura*^{null/+}, *sakura*^{null/null}, and *sakura-EGFP* rescue (*sakura*^{null/null} with *sakura-EGFP*) flies.

piRNA levels in *sakura*^{null/null} ovaries were reduced compared to controls (*sakura*^{+/+}), while *sakura*^{null/+} and *sakura-EGFP* rescue showed no such reduction (**Fig 4A**). Consistent with this, transposon RNA levels in *sakura*^{null/null} ovaries were upregulated compared to *sakura*^{+/+}, while *sakura*^{null/+} and *sakura-EGFP* rescue showed no such upregulation (**Fig 4B**). Given that Sakura is specifically expressed in germline cells, but not in somatic cells (**Fig 1E** and **1F**), we hypothesized that loss of *sakura* interrupts the piRNA pathway in germ cells, leading to the desilencing of germline transposons. A well-characterized germline transposon in *Drosophila* is *Burdock* (Donertas et al. 2013; Handler et al. 2013). We found that *Burdock* piRNA levels were significantly lower in *sakura*^{null/null} compared to controls and rescue flies (**Fig 4C**).

We employed the *Burdock* sensor, a transposon reporter tool to monitor germline piRNA activity (Handler et al. 2013). The reporter expresses nuclear GFP and β -gal under the control of the *nanos* promoter for germline expression, with a target sequence for *Burdock* piRNAs in the 3' UTR (**Fig 4D**). Using a *sakura* RNAi line driven by *UAS-Dcr2* and *NGT-Gal4* to knock down *sakura* specifically in the germline, we observed that GFP and β -gal expression were highly elevated in *sakura* knockdown germlines compared to control RNAi (*y^{RNAi}*) knockdowns (**Fig 4D**). This

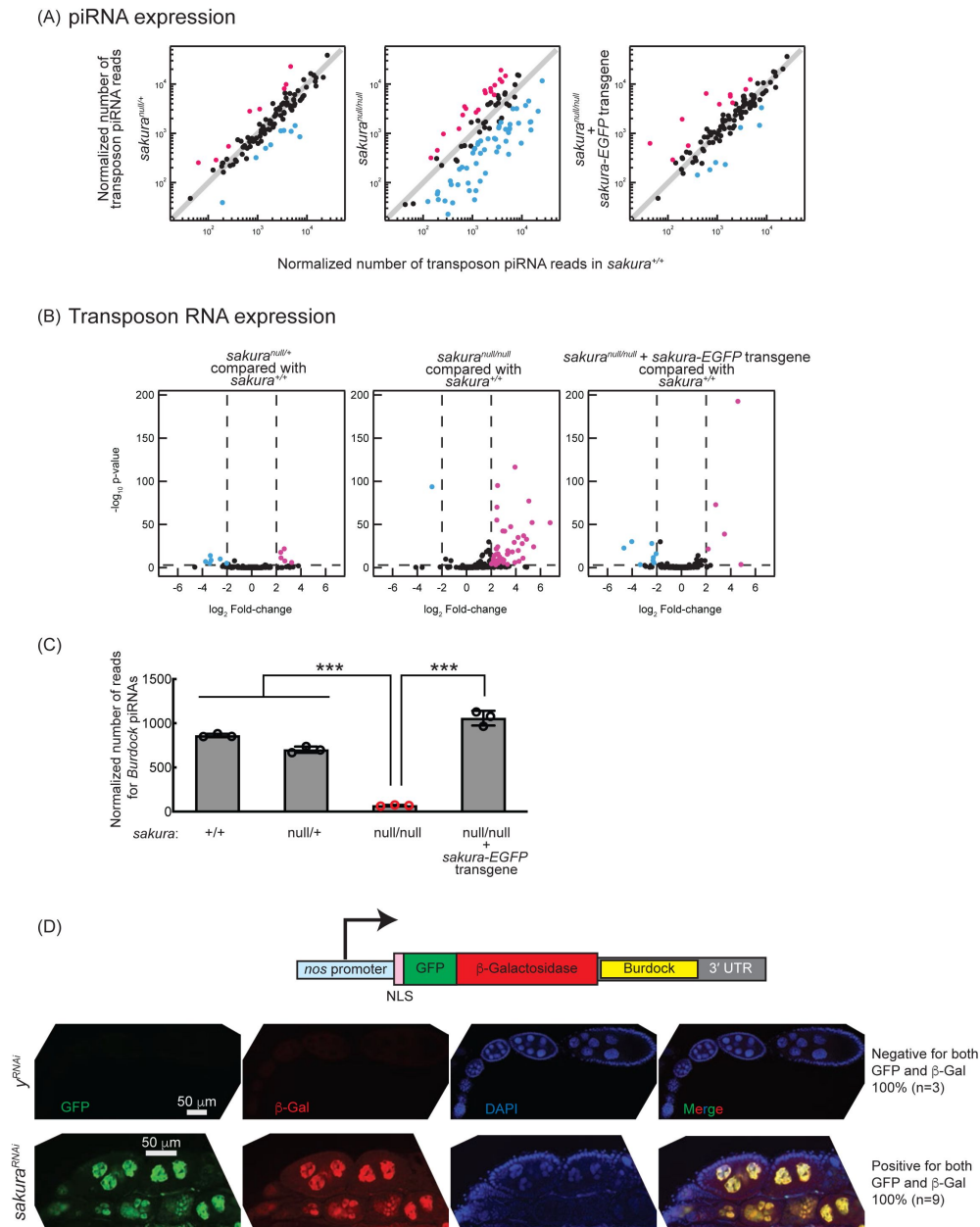


Fig 4.

Loss of *sakura* results in lower piRNA levels and loss of piRNA-mediated transposon silencing

(A) Scatter plots of normalized number of transposon piRNA reads from small RNA-seq of indicated genotypes compared with *sakura*^{+/+}. Means of three biological replicates are plotted. Downregulated (fold-change < 0.5) and upregulated (fold-change > 2) transposon piRNAs are shown in cyan and magenta, respectively. (B) Volcano plots of transposon RNAs from RNA-seq of indicated genotypes compared with *sakura*^{+/+}. Three biological replicates per genotype were analyzed. Downregulated (adjusted p-value < 0.001 and log₂(fold-change) < -2) and upregulated (adjusted p-value < 0.001 and log₂(fold-change) > 2) transposons are shown in cyan and magenta, respectively. (C) Normalized number of reads for *Burdock* piRNAs from small RNA-seq. Mean ± SD (n = 3). P-value < 0.001 (Student's t-test, unpaired, two-tailed) is indicated by ***. (D) The *Burdock* sensor harbors a *nanos* promoter, a nuclear localization signal (NLS) appended to GFP and β-gal coding sequences, and a target sequence for *Burdock* piRNAs in the 3'UTR. Confocal images of ovaries from control (*y*^{RNAi}) and *sakura*^{RNAi} flies harboring the *Burdock* sensor, where RNAi knockdown was specifically driven in the female germline with *UAS-Dcr2* and *NGT-Gal4*. GFP (green), β-gal (red), and DAPI (blue). Scale bars: 50 μm. Three out of 3 tested control samples were negative for both GFP and β-gal, while 9 out of 9 tested *sakura*^{RNAi} samples were positive for both markers.

confirmed that loss of *sakura* leads to a loss of piRNA-mediated transposon silencing in the germline, suggesting that Sakura is essential for proper piRNA levels and piRNA-based transposon silencing in the germline.

Sex-specific *sxl* mRNA alternative splicing

is dysregulated in *sakura*^{null/null} ovaries

Sxl is a master regulator of sex determination in *Drosophila* (Penalva and Sanchez 2003 [↗](#); Salz and Erickson 2010 [↗](#); Grmai et al. 2022 [↗](#)). Sex-specific alternative splicing of *sxl* transcripts produces distinct mRNA isoforms: the female-specific and male-specific mRNA isoforms in respective sex and only the female-specific mRNA isoform encodes the functional *Sxl* protein, whereas the male-specific isoform does not. In the female germline, loss of *Sxl* function or disruption of female-specific *sxl* splicing leads to developmental defects, including germline tumors and sterility. Notably, several mutants with tumorous ovariole phenotypes—including *otu* mutants—exhibit aberrant *sxl* mRNA splicing, resulting in the expression of the male-specific isoform in ovaries and defects in germ cell sexual identity (Bopp et al. 1993 [↗](#); Pauli et al. 1993 [↗](#)).

We examined *sxl* splicing in *sakura* mutants. As expected, ovaries from control and *sakura*-EGFP rescue flies expressed exclusively the female-specific *sxl* mRNA isoform and testes from control flies expressed only the male-specific isoform (Fig S6 [↗](#)). In contrast, *sakura*^{null/null} ovaries exhibited the male-specific isoform and a reduced level of the female-specific isoform. These results indicate that female-specific *sxl* alternative splicing is disrupted in the absence of *sakura*.

sakura is important for oogenesis in germline cells beyond GSCs and germline cysts

Both *sakura*^{null/null} and *sakura* germline RNAi knockdown driven by *UAS-Dcr-2* and *NGT-Gal4*, which initiates RNAi in germline from GSCs, resulted in rudimentary ovaries (Fig 2C [↗](#) and 5A [↗](#)). These ovaries lacked later-stage germline cells and egg chambers, making it difficult to assess the role of Sakura beyond the germarium. However, because *Sakura*-EGFP expression is not limited to GSCs and germline cysts (Fig 1E [↗](#) and 1F [↗](#)), we speculated that Sakura might function in later-stage germline cells as well.

To explore this possibility, we used *TOsk-Gal4* (a combination of *osk-Gal4* and *αTub67C-Gal4*) to drive *sakura* RNAi knockdown in germline cells after the germline cyst stage (from germarium region 2b onward. Fig 1A [↗](#)), sparing GSCs and germline cysts (ElMaghraby et al. 2022 [↗](#)). Unlike *NGT-Gal4*, *TOsk-Gal4*-driven *sakura* RNAi knockdown did not drastically affect ovary morphology (Fig 5A [↗](#)), allowing us to study the effects of *sakura* loss in egg chambers. We confirmed *sakura* RNAi (*sakura* RNAi #1 and #2) knockdown efficiency by Western blot, showing effective depletion of Sakura in ovaries from both *NGT-Gal4* and *TOsk-Gal4*-driven RNAi lines (Fig 5B [↗](#)).

Interestingly, *TOsk-Gal4*-driven *sakura* RNAi knockdown severely reduced the numbers of eggs laid (Fig 5C [↗](#)) and stage 14 oocytes in ovaries compared to control RNAi (*y*^{RNAi}) (Fig 5D [↗](#)), suggesting that *sakura* is important for oogenesis beyond the germline cyst stage. Additionally, we observed mislocalized and dispersed Oo18 RNA-binding protein (Orb), a marker for oocyte identity, in *TOsk-Gal4*-driven *sakura* RNAi ovaries beginning around stage 6 of oogenesis (Fig 5E [↗](#)). In controls, Orb was enriched and properly localized to the posterior end of stage ~6-8 egg chambers (Fig 5E [↗](#), yellow arrows). In *sakura*^{RNAi} ovaries, although Orb localization appeared normal at stage ~6 (yellow arrow), mislocalization was evident by stage ~8 (magenta arrow), and signs of cytoskeletal disorganization were apparent in later-stage egg chambers (white arrowheads). Phalloidin staining to visualize F-actin further supported cytoskeletal defects in *sakura*^{RNAi} later-stage egg chambers (Fig S7 [↗](#)).

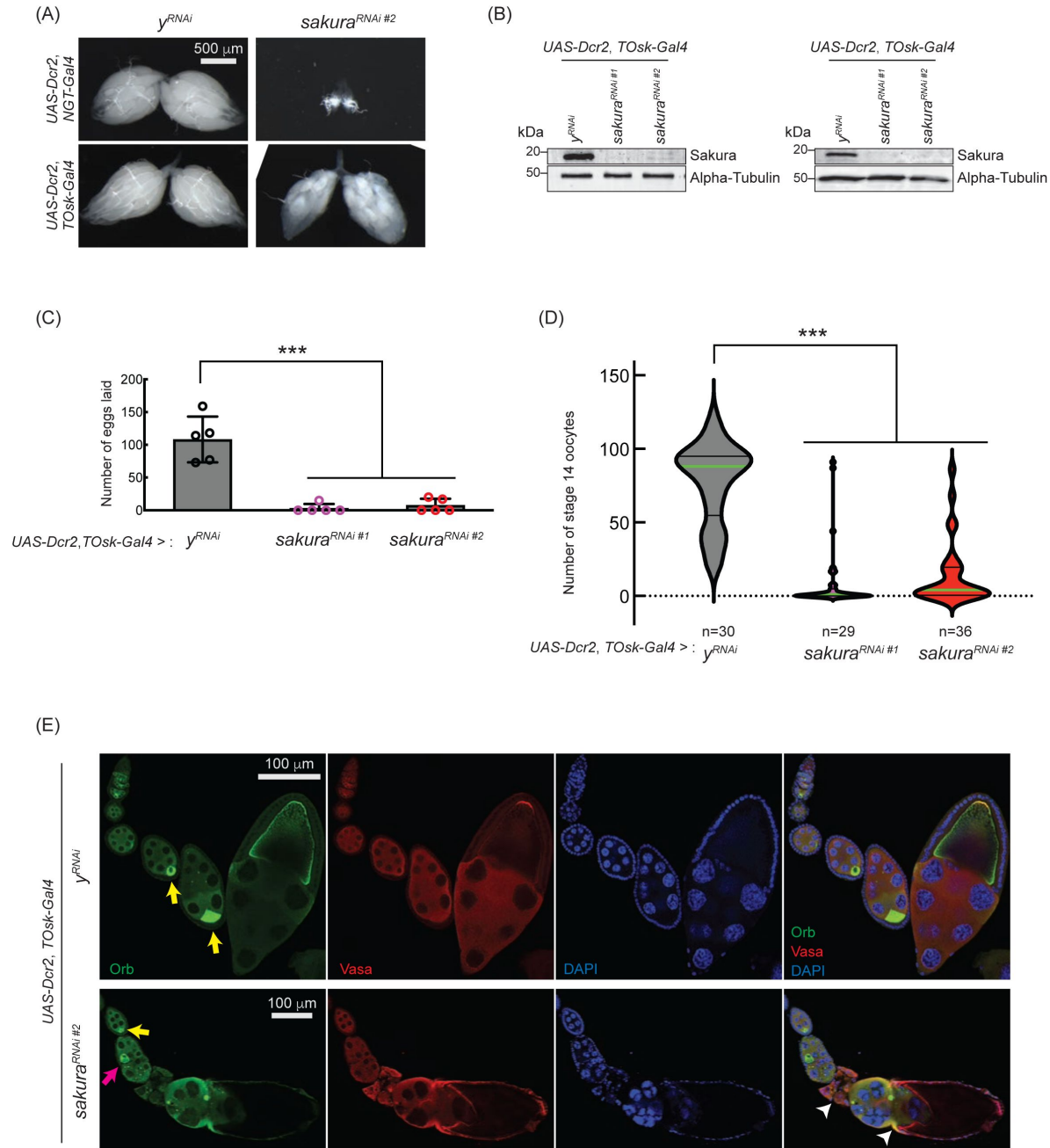


Fig 5.

***sakura* is important for oogenesis in later-stage egg chambers**

(A) Stereomicroscope images of dissected whole ovaries. Scale bar: 500 μ m. (B) Western blot of dissected Ovary lysates. *sakura^{RNAi} #1* and *sakura^{RNAi} #2* are two independent RNAi lines. (C) Number of eggs laid by *sakura* RNAi knockdown driven by *UAS-Dcr2* and *TOSk-Gal4*. Mean $\pm$ SD (n = 5). P-value < 0.001 (Student's t-test, unpaired, two-tailed) is indicated by ***. (D) Number of stage 14 oocytes produced by *sakura* RNAi knockdown flies driven by *UAS-Dcr2* and *TOSk-Gal4*. P-value < 0.001 (Student's t-test, unpaired, two-tailed) is indicated by ***. (E) Confocal images of the ovaries from *sakura* RNAi knockdown flies driven by *UAS-Dcr2* and *TOSk-Gal4*, stained with anti-Orb and anti-Vasa antibodies. Orb (green), Vasa (red), and DAPI (blue). Yellow arrows label the normal enrichment of Orb in the developing oocytes. Magenta arrow labels mislocalized developing oocyte. White arrowheads label egg chambers with cytoskeletal disorganization. Scale bars: 100 μ m. In A-E, *y^{RNAi}* was used as a control.

These defects likely contribute to the impaired production of stage 14 oocytes and eggs in *sakura*^{RNAi} flies (**Fig 5C** and **5D**). We conclude that Sakura plays an essential role in oogenesis beyond the early germline stages. For all subsequent *sakura* RNAi experiments, we used the *sakura* RNAi #2 line.

Sakura is required intrinsically for GSC establishment, maintenance, and differentiation

To investigate whether *sakura* functions autonomously in the germline, we performed mosaic analysis using the FLP-FRT system with heat shock promoter-driven FLP (*hs-flp*) (Xu and Rubin 1993; Rubin and Huynh 2015). First, to assess its role in GSC establishment, we induced *sakura*^{null} clones in primordial germ cells (PGCs) by applying heat shock before the early third instar larval stage and tracked their development into adult GSCs, following a previously established protocol (Yang et al. 2007). In control adult flies (*FRT82B*), 12.3% (20/163) of GSCs were marked, whereas only 1.9% (4/213) were marked in *sakura*^{null} mutants (*FRT82B, sakura*^{null}) (p-value = 4.8×10^{-11} , chi-square test), indicating that *sakura* is autonomously required for GSC establishment.

Next, we assessed GSC maintenance using the same system by generating *sakura*^{null} GSC clones in adult flies and measuring clone loss over time, as described previously (Xie and Spradling 1998; Yang et al. 2007). Four days after clone induction, 27.8% of GSCs were marked in controls (*FRT82B*) and 22.1% in *sakura*^{null} (*FRT82B, sakura*^{null}), which we defined as initial levels (**Fig 6A**). In controls, the percentage of marked GSCs declined to 25.8% on day 7 and 16.8% on day 14, resulting in a 39.4% loss rate over 10 days (from day 4 to day 14). In contrast, the proportion of marked *sakura*^{null} GSC clones dropped more sharply—to 16.3% on day 7 and 5.4% on day 14—yielding a higher loss rate of 75.6% over 10 days. These results demonstrate that *sakura* is intrinsically important for GSC maintenance.

We further tested whether *sakura* is intrinsically required for GSC differentiation, we quantified the number of marked (GFP-negative) and unmarked (GFP-positive) GSC-like cells (germline cells with a round spectrosome) in the germaria containing marked GSC clones at days 4, 7, and 14 post-induction (**Fig 6B**). We found that marked *sakura*^{null} GSC-like cells were significantly more numerous than marked control GSC-like cells at all time points (**Fig 6C**). The number of marked *sakura*^{null} GSC-like cells increased over the 10-day period, while the number of marked control cells did not (**Fig 6C**). In contrast, unmarked GSC-like cells in germaria containing *sakura*^{null} GSC clones did not differ significantly in number compared with those in germaria containing marked control GSC clones, and the number did not increase over time (**Fig S8**).

These results demonstrate that *sakura* is intrinsically important in germline cells, including GSCs, to regulate proper division and differentiation. In the absence of intrinsic *sakura*, germline cells become tumorous and undergo uncontrolled proliferation. Taken together, we conclude that *sakura* is required intrinsically for GSC establishment, maintenance, and differentiation.

Loss of *sakura* inhibits Dpp/BMP signaling

The Dpp/BMP signaling pathway plays a central role in regulating *bam* expression and GSC self-renewal and differentiation (**Fig S1**) (Kirilly and Xie 2007; Hayashi et al. 2020). We hypothesized that the germless and tumorous phenotypes observed in *sakura* loss-of-function ovaries could be due to Dpp/BMP signaling misregulation. To test this, we knocked down *sakura* in the germline driven by *UAS-Dcr2* and *NGT-Gal4* in the presence of *bam-GFP* reporter (Chen and McKearin 2003b). In controls (*y*^{RNAi}), Bam-GFP expression was restricted to 8-cell cysts and disappeared in 16-cell cysts and onward (**Fig 7A**). In contrast, *sakura* knockdowns ovaries showed persistent Bam-GFP expression throughout the germarium including the GSC niche region (**Fig 7A**).

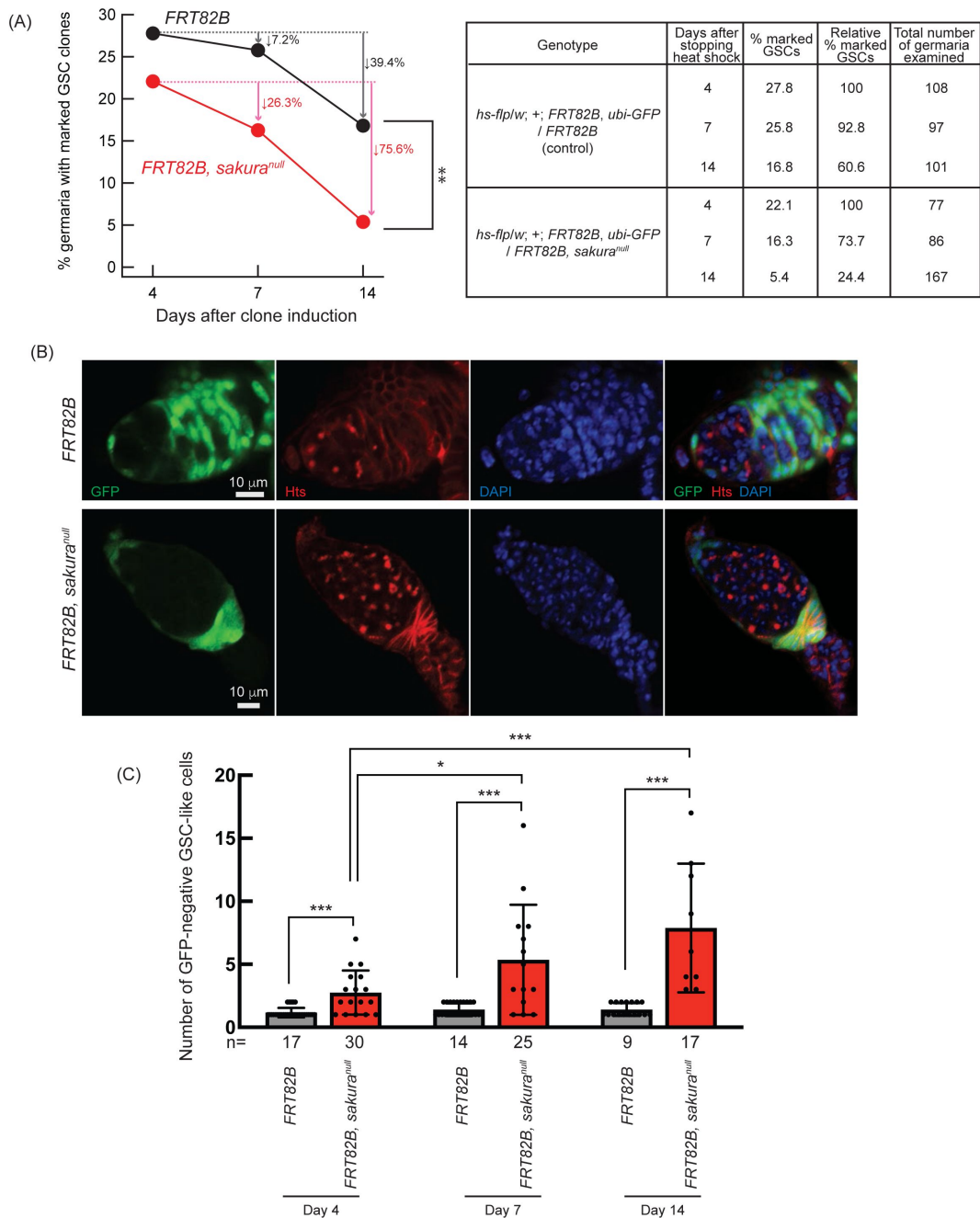


Fig 6.

Germline clonal analysis of *sakura^{null}*

(A) Percentage of germaria with marked GSC clones indicated by the absence of GFP at 4, 7, and 14 days after clone induction at the adult stage. Arrows indicate the percent decrease of marked GSC clones compared to day 4. The genotype, actual percentage, and total number of germaria examined are shown in the adjacent table. P-value < 0.01 (Chi-squared test) is indicated by **. (B) Confocal images of Germ cell clones. *sakura^{null}* and control clones were marked by the absence of GFP. GFP (green), Hts (red), and DAPI (blue). Scale bars: 10 μ m. (C) Number of marked (GFP-negative) GSC-like cells in germaria with marked GSCs of the indicated genotypes at 4, 7, and 14 days after clone induction. GSC-like cells containing round spectrosome were identified through immunostaining with anti-Hts antibody. P-value < 0.05 and <0.001 (Student's t-test, unpaired, two-tailed) is indicated by * and ***, respectively.

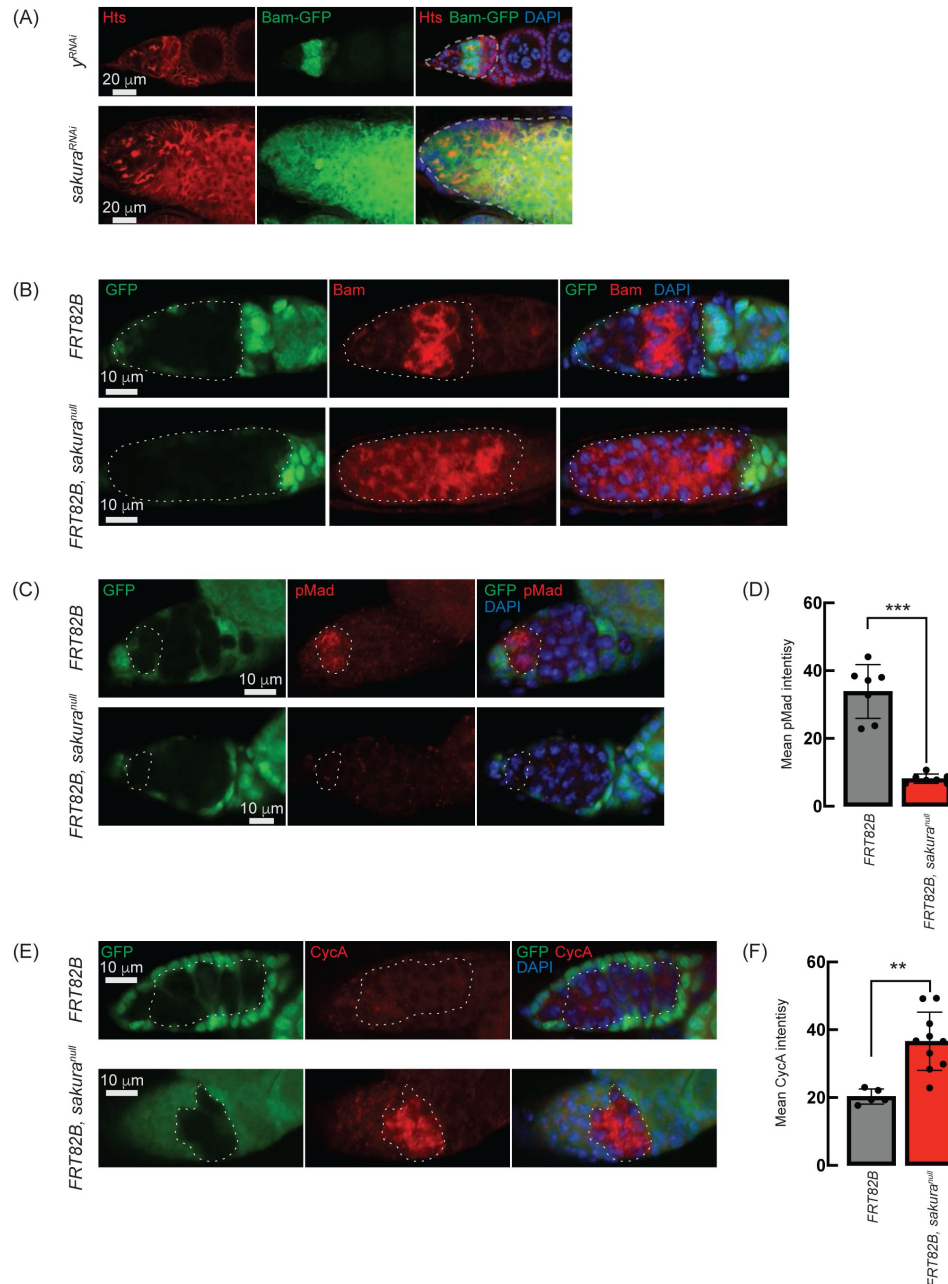


Fig 7.

Loss of *sakura* inhibits Dpp/BMP signaling

(A) Confocal images of ovaries from control (*y^{RNAi}*) and *sakura^{RNAi}* flies harboring the *bam-GFP* reporter, where RNAi knockdown was specifically driven in the female germline with *UAS-Dcr2* and *NGT-Gal4*. Bam-GFP (green), Hts (red), and DAPI (blue). Germariums are outlined by dotted line. Scale bars: 20 μ m. (B, C) Confocal images of germaria with germline clones of *sakura^{null}* stained with (B) anti-Bam antibody and (C) anti-pMad antibody. GFP (green), Bam or pMad (red), and DAPI (blue). Scale bars: 10 μ m. (D) Mean pMad intensity in the germline clones of the indicated genotypes. Mean $\pm$ SD ($n = 7$). P-value < 0.001 (Student's t-test, unpaired, two-tailed) is indicated by *** (E) Confocal images of germaria with germline clones of *sakura^{null}* stained with anti-CycA antibody. GFP (green), CycA (red), and DAPI (blue). Scale bars: 10 μ m. (F) Mean CycA intensity in the germline clones of the indicated genotypes. Mean $\pm$ SD ($n = 5$ and 10 for *FRT82B* and *FRT82B, sakura^{null}* respectively). P-value < 0.01 (Student's t-test, unpaired, two-tailed) is indicated by **. In B, C, and E, clones were marked with the absence of GFP.

To further confirm this finding, we used the FLP-FRT system to induce *sakura*^{null} clones in germaria and performed immunostaining with anti-Bam antibody. In control clones (*FRT82B*), Bam expression was observed exclusively in 8-cell cysts. However, in all *sakura*^{null} clones (*FRT82B*, *sakura*^{null}), Bam was aberrantly expressed throughout the germarium, including in GSCs (**Fig 7B**). These results suggest that in the absence of *sakura*, Bam expression is no longer repressed by Dpp/BMP signaling in the GSC niche, leading to GSC loss, and is no longer shut off after the 16-cell cyst stage.

In GSCs, pMad translocates into the nucleus to repress Bam expression (**Fig S1**) (Kai and Spradling 2003; Song et al. 2004). We stained ovaries with anti-pMad antibodies and found that pMad intensity was significantly reduced in *sakura*^{null} GSC clones compared to control GSCs (**Fig 7C and D**). This indicates that the dysregulated Bam expression observed in *sakura*^{null} was due to reduced pMad levels in GSCs.

Previous studies have shown that ectopic expression of a stable form of CycA leads to germ cell loss (Chen et al. 2009). This germ cell loss phenotype is also observed upon ectopic *bam* expression in GSCs (Xie and Spradling 1998; Chen and McKearin 2003a; Xia et al. 2010). It was reported that Bam associates with Ovarian tumor (Otu) to promote deubiquitination and stabilization of CycA (Ji et al. 2017). Since we observed derepressed *bam* expression in *sakura*^{null} cells, we investigated whether CycA levels were affected. Staining ovaries with anti-CycA antibodies revealed higher CycA intensity in *sakura*^{null} clones compared to neighboring wild-type cells in the same germarium (*FRT82B*, *sakura*^{null}) and control clones in control germarium (*FRT82B*) (**Fig 7E**). *sakura*^{null} clones exhibited significantly higher mean CycA intensity compared to control clones (**Fig 7F**). The elevated CycA levels in *sakura*^{null} clones likely results from Bam misexpression and Bam-mediated stabilization.

To determine whether *bam* misexpression occurs in egg chambers in *sakura* knockdown conditions, we examined *bam* expression in TOsk-Gal4-driven *sakura* RNAi flies. Bam was appropriately restricted to 8-cell cysts in germaria and not misexpressed in egg chambers compared to yRNAi controls (Fig. S9, cyan arrowheads). Thus, *sakura* is specifically required in GSCs and cyst cells to regulate *bam* expression.

We also examined DE-Cadherin (DE-Cad), which plays a role in oocyte positioning within the egg chamber (Godt and Tepass 1998). DE-Cad level and localization appeared normal in both control and *sakura* RNAi ovaries through approximately stage 8. However, later-stage egg chambers in *sakura* RNAi flies displayed structural disorganization (Fig. S9, white arrowheads), consistent with defects observed in **Fig. 5E** and Fig. S7.

Attempts to rescue *sakura* loss-of-function ovariole phenotypes

To determine whether the phenotypes caused by *sakura* loss-of-function could be rescued, we first performed genetic interaction experiments by knocking down either *bam* or *cycA* in the germline. As expected, germline-specific *bam* RNAi driven by *UAS-Dcr2* and *NGT-Gal4* resulted in 100% tumorous ovarioles, while *cycA* RNAi produced 100% normal ovarioles in 2–5-day-old flies (Fig. S10A). Double knockdown of *sakura* and *bam* reduced the proportion of germless ovarioles and increased the proportion of tumorous ovarioles compared to *sakura* RNAi alone or the double knockdown of *sakura* and control *w* RNAi. However, no normal ovarioles were observed (Fig. S10A). *sakura* and *cycA* double knockdown had no effect on the phenotype distribution, suggesting that the germless phenotype in *sakura* mutants is partially attributable to ectopic *bam* expression but not to *cycA* misregulation.

The number of GSC-like cells was increased in *sakura* and *bam* double knockdowns compared to *sakura* RNAi or *sakura* and *w* double RNAi, whereas *sakura* and *cycA* knockdown did not alter GSC-like cell numbers (Fig. S10B), further supporting a genetic interaction between *sakura* and *bam*.

Next, we tested whether the *sakura* loss-of-function phenotypes could be rescued by overexpressing components of the Dpp/BMP signaling pathway. Germline expression of *UASp-Mad-GFP* driven by *nos-Gal4-VP16* in both *sakura*^{null/+} controls and *sakura*^{null/null} did not alter the proportions of ovariole phenotypes (**Fig S11A**) or the number of GSC-like cells (**Fig S11B**). Thus, transgenic overexpression of Mad did not rescue the *sakura* mutant phenotypes.

We then tested whether overexpression of a constitutively active form of the Dpp receptor Thickveins (Tkv.Q253D) (Casanueva and Ferguson 2004) could rescue the phenotypes. Expression of *UASp-tnv.Q253D* in the germline using a *NGT-Gal4* in a control (*y*^{RNAi}) background resulted in 100% tumorous ovarioles, as expected (Fig. S12A). When co-expressed with *sakura* RNAi, Tkv.Q253D increased the proportion of tumorous ovarioles and decreased the proportion of germless ovarioles relative to *sakura*^{RNAi} alone but did not restore normal ovarioles. The number of GSC-like cells was also increased by Tkv.Q253D expression in *sakura*^{RNAi} (**Fig S12B**), indicating partial phenotypic modulation via BMP pathway activation.

Sakura binds Otu

To explore the molecular function of Sakura in oogenesis, we aimed to identify Sakura-interacting proteins. We conducted a co-immunoprecipitation experiment using anti-GFP magnetic beads on the ovary lysates of flies expressing Sakura-EGFP, followed by mass spectrometry to identify co-immunoprecipitated proteins. Ovary lysates from flies without the *sakura-EGFP* transgene (*w1118*) served as negative controls. Mass spectrometric analysis revealed several proteins with peptide signals present in all three biological replicates of the Sakura-EGFP samples without any signals in any of the three biological replicates of the negative control (**Table 1**). Among these, the most promising candidate, exhibiting the highest unique peptide signal in each of three replicates of the Sakura-EGFP samples was Otu.

To confirm this finding, we generated a polyclonal anti-Otu antibody and validated it by Western blots using ovary lysates from the hypomorphic *otu*¹⁴ mutant (**Fig S13A**) (King et al. 1986; Storto and King 1987; Steinhauer and Kalfayan 1992a; Pauli et al. 1993). Using this antibody, we detected endogenous Otu protein in the Sakura-EGFP immunoprecipitants from ovaries of 3–7-day-old flies, but not in the negative control (**Fig 8A**), validating the mass spectrometry results. Next, using wild-type (without Sakura-EGFP) ovary lysates from 3–7-day old-flies, we immunoprecipitated endogenous Sakura with anti-Sakura antibody and detected endogenous Otu in the immunoprecipitant by Western blot (**Fig 8B**), demonstrating that endogenous Sakura interacts with endogenous Otu. Furthermore, we found that ectopically expressed Flag-tagged Otu co-immunoprecipitated with ectopically expressed HA-tagged Sakura in S2 cells (**Fig 8C**).

The *otu* gene encodes two annotated protein isoforms, a predominant 98 kDa isoform and a less abundant 104 kDa isoform generated by alternative splicing (Steinhauer and Kalfayan 1992b). The 98 kDa isoform lacks the Tudor domain, which is encoded by an alternatively spliced 126-nucleotide exon (**Fig 9B**, Otu(ΔTudor)) (Van Buskirk and Schupbach 2002). Previous studies have shown that the 104 kDa Otu isoform is more abundant in predifferentiated germline cells and is sufficient to carry all known Otu functions, whereas the 98 kDa Otu isoform becomes prominent during later stages of oogenesis and is implicated in nurse cell regression and oocyte maturation (Sass et al. 1995). Consistent with these findings, we observed that ovaries from very young (2–5 hour old) flies—containing only germaria and previtellogenic egg chambers—predominantly expressed the 104 kDa isoform (**Fig S13B**). In contrast, ovaries from 3–7 day old flies—containing all 14 stages of oogenesis—predominantly expressed the 98 kDa isoform. Importantly, both the 104 kDa and 98 kDa isoforms co-immunoprecipitated with endogenous Sakura using anti-Sakura antibody, demonstrating that Sakura interacts with both isoforms of Otu and that the Tudor domain is not required for this interaction (Fig. S13B).

Table 1.

Number of unique peptide counts detected by mass-spec

. Proteins with peptide signals present in all three biological replicates of the Sakura-EGFP samples without any signals in any of the three biological replicates of the negative control are shown.

Identified protein name	Gene ID	Unique peptide counts					
		Sakura-EGFP samples			w1118 samples (negative control)		
		Replicate 1	Replicate 2	Replicate 3	Replicate 1	Replicate 2	Replicate 3
Ovarian tumor	CG12743	44	38	18	0	0	0
Uncharacterized protein Dmel_CG4679	CG4679	17	8	2	0	0	0
Uncharacterized protein Dmel_CG14997	CG14997	17	7	2	0	0	0
Mitochondrial ribosomal protein S22	CG12261	14	6	1	0	0	0
Tudor	CG9450	15	1	1	0	0	0
Mitochondrial ribosomal protein S5	CG40049	10	4	1	0	0	0
HECT and RLD domain containing protein 2	CG11734	4	9	1	0	0	0
Mitochondrial ribosomal protein S10	CG4247	5	3	1	0	0	0
Mitochondrial ribosomal protein S31	CG5904	3	1	1	0	0	0
Uncharacterized protein Dmel_CG1316	CG1316	2	1	1	0	0	0

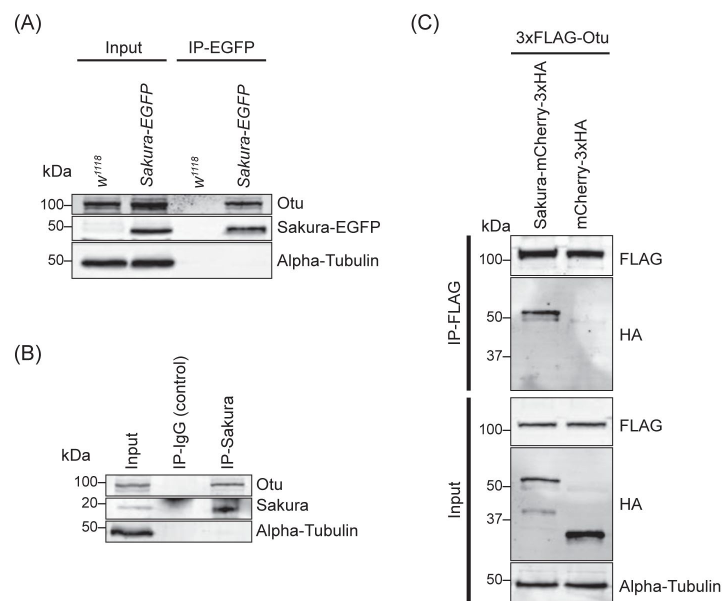


Fig 8.

Sakura interacts with Otu.

(A) Co-immunoprecipitation using anti-GFP magnetic beads followed by Western blotting. Ovary lysates expressing Sakura-EGFP in *sakura*^{+/+} background and those from *w*¹¹¹⁸ negative control were tested. (B) Co-immunoprecipitation using beads bound with rabbit anti-Sakura followed by Western blotting. Ovary lysates from *w*¹¹¹⁸ flies were used. Rabbit IgG was used as control IP. (C) Co-immunoprecipitation using beads bound with anti-FLAG antibody followed by Western blotting. S2 cell lysates expressing 3xFLAG-Otu and Sakura-mCherry-3xHA or mCherry-3xHA (negative control) were used.

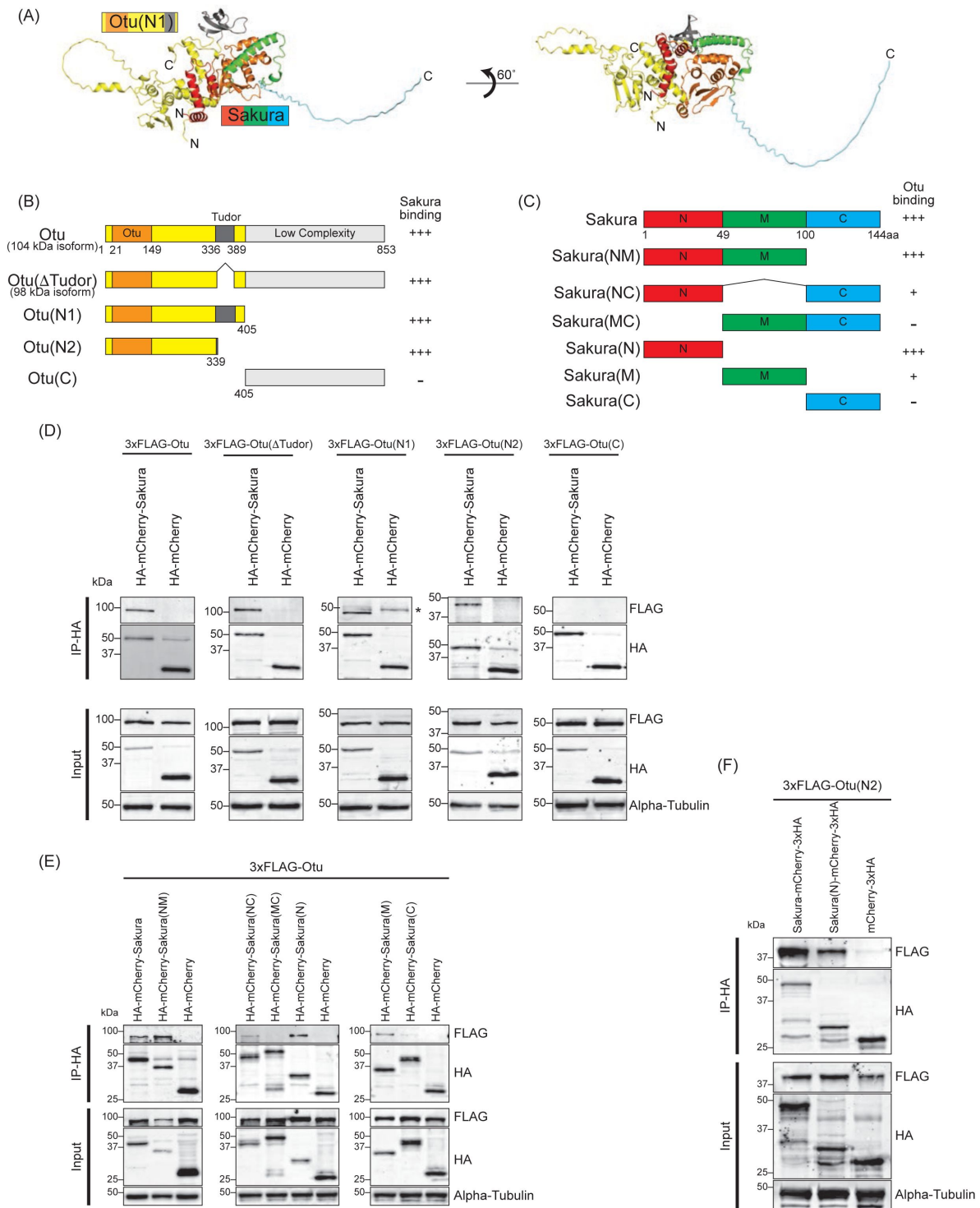


Fig 9.

N-terminal regions of Sakura and Otu are important for interaction.

(A) Predicted structure of the Sakura and Otu protein complex made by AlphaFold. Full-length Sakura and N-terminal Otu fragment (N1, 1-405aa) were used for prediction. (B) Full-length Otu and Otu fragments tested in co-immunoprecipitation assays. The binding assay results from (D) is summarized. (C) Full-length Sakura and Sakura fragments tested in co-immunoprecipitation assays (N: N-terminal, M: middle, C: C-terminal). The binding assay results from (E) is summarized. (D, E, F) Co-immunoprecipitation using anti-HA magnetic beads followed by Western blotting. S2 cell lysates expressing HA-tagged mCherry were used as negative controls.

Next, we sought to identify which regions of Sakura and Otu are important for their interaction. We predicted the structure of the Sakura-Otu protein complex using AlphaFold (Jumper et al. 2021 [↗](#); Yang et al. 2023 [↗](#)). AlphaFold suggested that the N-terminal region of Sakura (1-49aa) and the N-terminal region of Otu (1-405aa) are highly likely to form interactions, while the C-terminal region of Sakura (100-144aa) does not interact with Otu (**Fig 9A-C** [↗](#)). The Tudor domain of Otu is not directly involved in this interaction, consistent with our findings (**Fig S13B** [↗](#)). To experimentally validate the predictions made by AlphaFold, we generated a series of truncated Otu fragments, including Otu(Δ Tudor) (1-336, 389-853aa, corresponding to the endogenous 98 kDa isoform), Otu(N1) (1-405aa), Otu(N2) (1-339aa), and Otu(C) (405-853aa) (**Fig 9B** [↗](#)). We then performed co-immunoprecipitation assays to test which fragments could interact with full-length Sakura. All fragments except Otu(C) were associated with full-length Sakura protein in S2 cells (**Fig 9D** [↗](#)), demonstrating that the C-terminal low-complexity region of Otu is dispensable for the interaction. This assay also showed that the Tudor domain is not critical and the N-terminal fragment (1-339aa) of Otu is sufficient for interaction with Sakura. These findings align well with the predictions made by AlphaFold.

Subsequently, we created several truncated Sakura fragments and conducted co-immunoprecipitation assays in S2 cells to determine which Sakura fragments can interact with full-length Otu. The fragments generated included Sakura(NM) (1-100aa), Sakura(NC) (1-49, 100-144aa), Sakura(MC) (49-144aa), Sakura(N) (1-49aa), Sakura(M) (49-100aa), and Sakura(C) (100-144aa) (**Fig 9C** [↗](#)). We found that Sakura(NM), Sakura(NC), Sakura(N), and Sakura(M) were associated with Otu (**Fig 9E** [↗](#)). Compared to Sakura(NM) and Sakura(N), the interaction of Sakura(M) with Otu is relatively weaker, as indicated by a fainter band in the Western blot (**Fig 9E** [↗](#)). Sakura-NC exhibited a weak interaction with Otu, while Sakura(MC) and Sakura(C) showed no interaction (**Fig 9E** [↗](#)). These assays demonstrated that the N-terminal region (1-49aa) of Sakura is sufficient for interaction with Otu. Moreover, the results suggest that the C-terminal region of Sakura (100-144aa) does not interact with Otu and that adjoining it with the N-terminal (1-49aa) or the middle (M) region (49-100aa) weakens the interaction with Otu. These findings support the AlphaFold predictions.

Finally, to test whether the interaction between Sakura and Otu can be achieved solely through their N-terminal regions, we performed co-immunoprecipitation assay with Sakura(N) and Otu(N2). We discovered that Sakura(N) and Otu(N2) physically interact (**Fig 9F** [↗](#)). A reciprocal co-immunoprecipitation assay also confirmed their interaction (**Fig S14** [↗](#)), revealing that the interaction between Sakura and Otu can be established with just the 1-49aa of Sakura and the 1-339aa of Otu. Therefore, we conclude that Sakura interacts with Otu *in vivo*, and the N-terminal regions of both proteins are sufficient for their interaction.

Loss of *otu* phenocopies loss of *sakura*

Having established the interaction between Sakura and Otu, we were interested in whether they function together during oogenesis. We speculated that for them to interact and function together, Sakura and Otu must exhibit similar expression and localization patterns in ovaries. We generated two transgenic flies carrying full-length *otu-EGFP* or *otu(Δ Tudor)-EGFP* transgenes; both constructs are C-terminally fused with EGFP and under the control of the *otu* promoter. We found that both Otu-EGFP and Otu(Δ Tudor)-EGFP display similar expression and localization patterns to Sakura-EGFP (**Fig 1E** [↗](#), 1F and **Fig 10A** [↗](#)). Like Sakura-EGFP, Otu-EGFP and Otu(Δ Tudor)-EGFP are specifically expressed in germ cells, but not in follicle cells, and are localized to the cytoplasm, being enriched in the developing oocytes (**Fig 10A** [↗](#)). We did not observe any difference in localization patterns between Otu-EGFP and Otu(Δ Tudor)-EGFP; suggesting that the Tudor domain is not crucial for Otu protein localization. Notably, the localization pattern exhibited by the transgenic Otu-EGFP generated in this study is consistent with a previous report (Glenn and Searles 2001 [↗](#)).

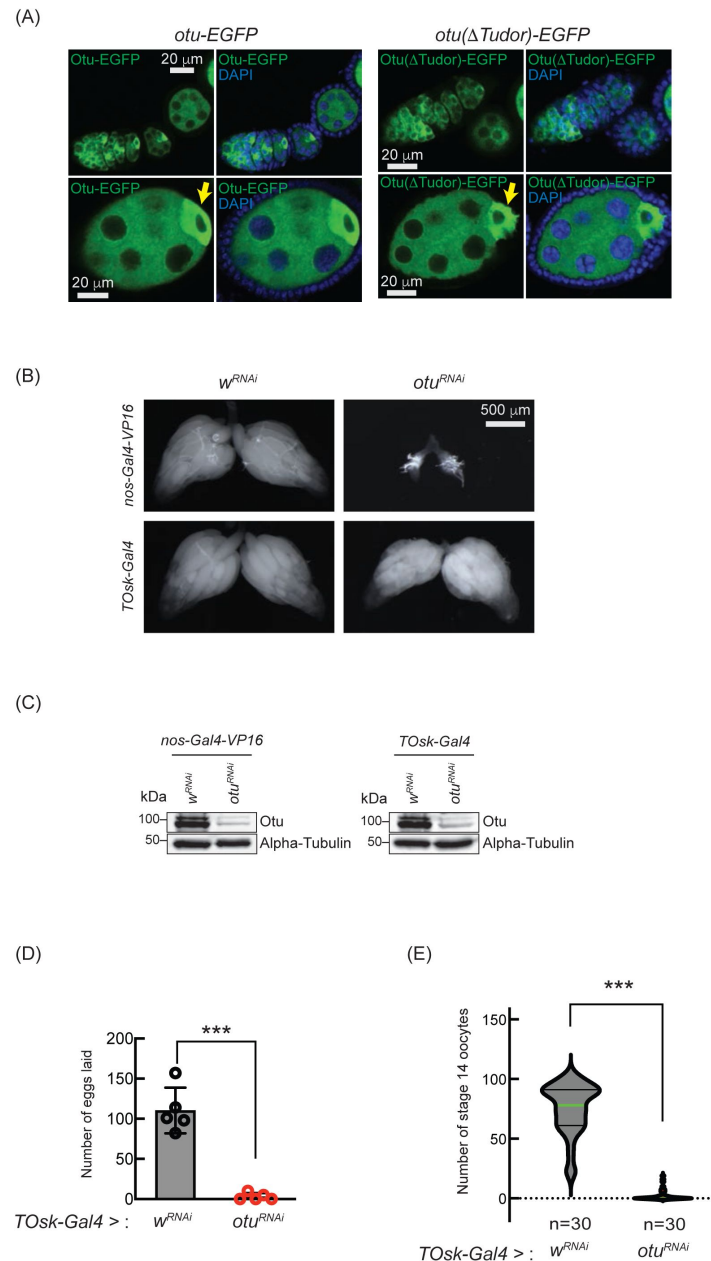


Fig 10.

Loss of *otu* phenocopies loss of *sakura*

(A) Confocal images of ovaries from *otu-EGFP* and *otu(ΔTudor)-EGFP* transgenic flies. Otu-EGFP and Otu(ΔTudor)-EGFP (green), DAPI (blue). Yellow arrows show the enrichment of Otu-EGFP and Otu(ΔTudor)-EGFP in developing oocytes. Scale bars: 20 μm. (B) Stereomicroscope images of dissected whole ovaries from *w^{RNAi}* (control) and *otu^{RNAi}* flies where RNAi knockdown was driven in the female germline with *nos-Gal4-VP16* or *TOsk-Gal4*. Scale bar: 500 μm. (C) Western blot of dissected ovary lysates. (D) Number of eggs laid by *TOsk-Gal4 > w^{RNAi}* and *TOsk-Gal4 > otu^{RNAi}* flies. Mean ± SD (n = 5). P-value < 0.001 (Student's t-test, unpaired, two-tailed) are indicated by ***. (E) Violin plots of the number of stage 14 oocytes produced in *TOsk-Gal4 > w^{RNAi}* and *TOsk-Gal4 > otu^{RNAi}* flies. n = 30. P-value < 0.001 (Student's t-test, unpaired, two-tailed) is indicated by ***.

Previous studies have shown that mutations in *otu* lead to defects in germ cell division and differentiation, resulting in phenotypes including tumorous ovarioles (King and Riley 1982 [↗](#); King et al. 1986 [↗](#); Storto and King 1988 [↗](#); Steinhauer and Kalfayan 1992b [↗](#)). To study the role of *otu* specifically in the germline, we performed RNAi knockdown of *otu* in the germline driven by *nos-Gal4-VP16* (Fig 10B [↗](#) and 10C [↗](#)). Interestingly, we found that germline depletion of *otu* results in rudimentary ovaries, similar to the loss of *sakura* (Fig 2C [↗](#), Fig 5A [↗](#), and Fig 10B [↗](#)). Furthermore, similar to *sakura* RNAi (Fig 5 [↗](#)), *TOsk-Gal4*-driven *otu* RNAi knockdown did not affect ovary morphology (Fig 10B [↗](#)), but it resulted in a reduced number of eggs laid and stage 14 oocytes compared with control RNAi (*w^{RNAi}*) (Figure 10D [↗](#) and 10E [↗](#)).

Given that *sakura* loss of function causes tumorous and germless ovarioles, apoptosis, piRNA pathway defects, *bam* misexpression, and reduced pMad levels (Fig 3 [↗](#), 4 and 7A-D), we asked whether similar defects occur upon *otu* depletion. Germline *otu* RNAi knockdown in 2-5 days old flies using *UAS-Dcr2* and *NGT-Gal4* lead to both germless (~70%) and tumorous (~30%) ovarioles, closely resembling the phenotype of *sakura* RNAi (Fig S10A [↗](#)). Similar germless and tumorous phenotypes were also observed in *nos-Gal4-VP16 > otu^{RNAi}* flies (Fig 11A [↗](#)). In these *otu^{RNAi}* ovarioles, Bam-GFP expression was no longer restricted to 8-cell cysts, instead persisting throughout the germarium and egg chambers (Fig 11A [↗](#)). Additionally, *otu^{14/14}* mutant GSCs showed reduced pMad levels (Fig 11B [↗](#)), and cleaved Caspase-3 staining revealed elevated apoptosis in *otu* RNAi ovaries (Fig 11C [↗](#)). Germline depletion of *otu* resulted in elevated expression of GFP and β -Gal produced by the *Burdock* sensor, indicating loss of piRNA-mediated transposon silencing (Fig 11D [↗](#)).

Together, these data demonstrate that *otu* loss-of-function closely phenocopies *sakura* loss-of-function, including defects in germline maintenance and differentiation, *bam* misexpression, reduced pMad signaling, and compromised piRNA pathway activity. These findings, along with physical interaction between Sakura and Otu (Figs 8 [↗](#), 9, S13, and S14), suggest that they function together in regulating germline maintenance and differentiation.

To further explore their genetic interaction, we compared the ovariole phenotypes caused by single and double knockdowns of *sakura* and *otu* in the germline driven by *UAS-Dcr2* and *NGT-Gal4*. Individual knockdown of either gene resulted in ~70% germless and ~30% tumorous ovarioles in 2-5 days old flies (Fig S10A [↗](#)). Simultaneous knockdown of both genes increased the proportion of germless ovarioles to 86%, with a corresponding decrease in tumorous phenotypes to 14%, suggesting a synergistic enhancement of the germless phenotype. The number of GSC-like cells were not affected by the double knockdown of *sakura* and *otu* compared with their respective single knockdown (Fig S10B [↗](#)).

Sakura and Otu proteins do not depend on each other for their abundance

We investigated whether Sakura and Otu proteins depend on each other for their abundance. We performed Western blot analysis on ovaries with germline-specific knockdown of *sakura* or *otu*, driven by *TOsk-Gal4*. Otu protein levels were not reduced in *sakura^{RNAi}* ovaries compared to control *y^{RNAi}* (Fig S15A [↗](#)), and conversely, Sakura levels were not decreased in *otu^{RNAi}* ovaries compared with control *w^{RNAi}* ovaries (Fig S15B [↗](#)). These results indicate that Sakura and Otu do not require each other for their protein expression or stability.

To further test this conclusion and to examine whether Otu enrichment to the posterior of egg chambers is affected by loss of *sakura*, we generated marked *sakura^{null}* germline clones using the FLP-FRT system. Clones were marked by the absence of RFP, and experiments were performed in the flies expressing the *otu-EGFP* or *otu(Δ Tudor)-EGFP* transgene in an otherwise wild-type (*otu^{+/+}*) background. Otu-EGFP fluorescence signal was comparable between *sakura^{null}* clones (*FRT82B*,

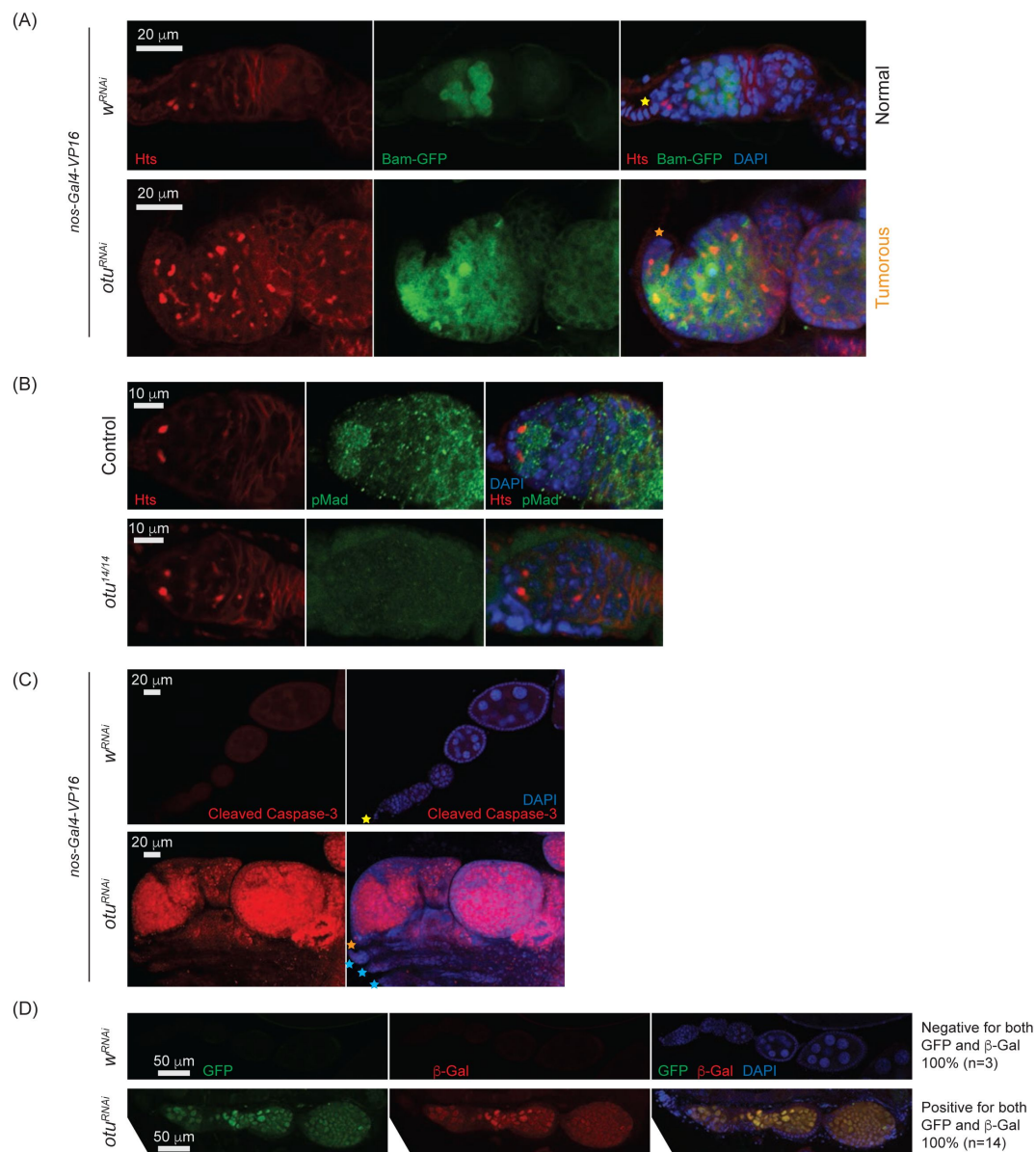


Fig 11.

Loss of *otu* results in low levels of pMad and derepression of *bam* in the germlaria

(A) Confocal images of the ovaries from control (w^{RNAi}) and otu^{RNAi} flies harboring the *bam*-GFP reporter, where RNAi knockdown was driven in the female germline with *nos-Gal4-VP16*. Bam-GFP (green), Hts (red), and DAPI (blue). Scale bars: 20 μm . (B) Confocal images of germaria from control ($otu^{14/+}$) and $otu^{14/14}$ mutant flies stained with anti-pMad (green) and anti-Hts (red) antibodies. Scale bars: 10 μm . (C) Confocal images of the ovaries from $nos-Gal4-VP16 > w^{RNAi}$ and $nos-Gal4-VP16 > otu^{RNAi}$ flies stained with anti-cleaved Caspase 3 antibody. Cleaved Caspase-3 (red) and DAPI (blue). Scale bars: 20 μm . (D) Confocal images of the ovaries from w^{RNAi} and otu^{RNAi} flies carrying the Burdock sensor, with RNAi driven in the female germline using *UAS-Dcr2*, *NGT-Gal4* and *nos-Gal4-VP16*. GFP (green), β -gal (red), and DAPI (blue). Scale bars: 50 μm . Three out of 3 tested control samples were negative for both GFP and β -gal, while 14 out of 14 tested otu^{RNAi} samples were positive for positive for GFP and β -gal.

sakura^{null}) and control clones (*FRT82B*) (Fig 12A). Similarly, Otu(ΔTudor)-EGFP levels were unaffected in *sakura*^{null} clones compared to the control clones (Fig 12B), confirming that Sakura is not required for Otu protein expression or stability.

Both Otu-EGFP and Otu(ΔTudor)-EGFP exhibited clear enrichment at the posterior of egg chambers in the control clones (*FRT82B*)—consistent with localization to the developing oocyte (Fig 12A and 12B). This posterior enrichment was absent in *sakura*^{null} clones, demonstrating that Sakura is crucial for this process.

Sakura does not affect Otu's deubiquitinase activity in vitro

A previous study has shown that Otu possesses deubiquitinase activity (Ji et al. 2017). We asked whether Sakura regulates the Otu's deubiquitinase activity. To test this, we performed an in vitro deubiquitination assay using Ub-Rhodamine 110 as a model substrate. Consistent with the previous study (Ji et al. 2017), we detected a deubiquitinase activity in Otu that was ectopically expressed and purified from S2 cells (Fig S16). The presence of Sakura, purified from *E.coli*, did not affect Otu's deubiquitinase activity under these assay conditions. Additionally, Sakura itself did not exhibit deubiquitinase activity.

Discussion

In this study, we identified Sakura, encoded by a previously uncharacterized gene *CG14545*, as an essential factor for oogenesis and female fertility. We demonstrated that Sakura is specifically expressed in germline cells in the ovary, including GSCs, is localized to the cytoplasm, and is enriched in the developing oocytes. *sakura* homozygous null mutant flies are viable but completely female-sterile and male-fertile. Loss of *sakura*, either through null mutation or germline RNAi including in GSCs, results in rudimentary ovaries exhibiting germless and tumorous phenotypes.

The tumorous phenotype associated with the loss of *sakura* is characterized by an excess of GSC-like cells, which feature round spectrosomes, as well as cyst cells with branched fusomes (Fig 3C). The increased number of GSC-like cells in loss of *sakura* suggests dysregulation of GSC self-renewal and differentiation. Meanwhile, the presence of excess cyst cells with branched fusomes indicates abnormal differentiation and division of cysts. Differentiation of cystoblasts typically involves four mitotic divisions with incomplete cytokinesis, leading to 16 cyst cells interconnected by branched fusomes. The degree of fusome branching serves as a marker for the stages of cyst cell division, with increased branching from 2-cell to 16-cell cysts (de Cuevas and Spradling 1998). Notably, fusomes begin to degenerate and disappear after 16-cell cysts as the germline cyst enters the meiotic zone or region 2 of the germarium. The persistence of cyst cells with branched fusomes in *sakura*^{null/null} suggests that *sakura* is crucial for the proper division and differentiation of cysts, and we speculate that it is required for germline cysts to enter meiotic division.

Mosaic analysis of *sakura*^{null} indicates that *sakura* is intrinsically required for the establishment of GSCs in the ovary. When mutant clones were induced in PGC stage, significantly fewer *sakura*^{null} marked GSCs were observed in adult ovaries, suggesting that many *sakura*^{null} PGCs fail to survive or differentiate into GSCs. Furthermore, induction of *sakura*^{null} clones in adult ovaries led to a more rapid decline in marked *sakura*^{null} GSCs compared to controls, indicating that *sakura* is also intrinsically required for GSC maintenance (Fig 6A). Over time, germaria containing *sakura*^{null} GSC clones became increasingly tumorous, with an expanding population of *sakura*^{null} GSC-like cells (Fig 6B and 6C). In contrast, the number of GSC-like cells across entire and all ovarioles regardless of whether GSCs in the niche remain, declined over time in *sakura*^{null} ovaries (Fig S5B). These results suggest that *sakura*^{null} GSCs initially undergo aberrant proliferation, leading

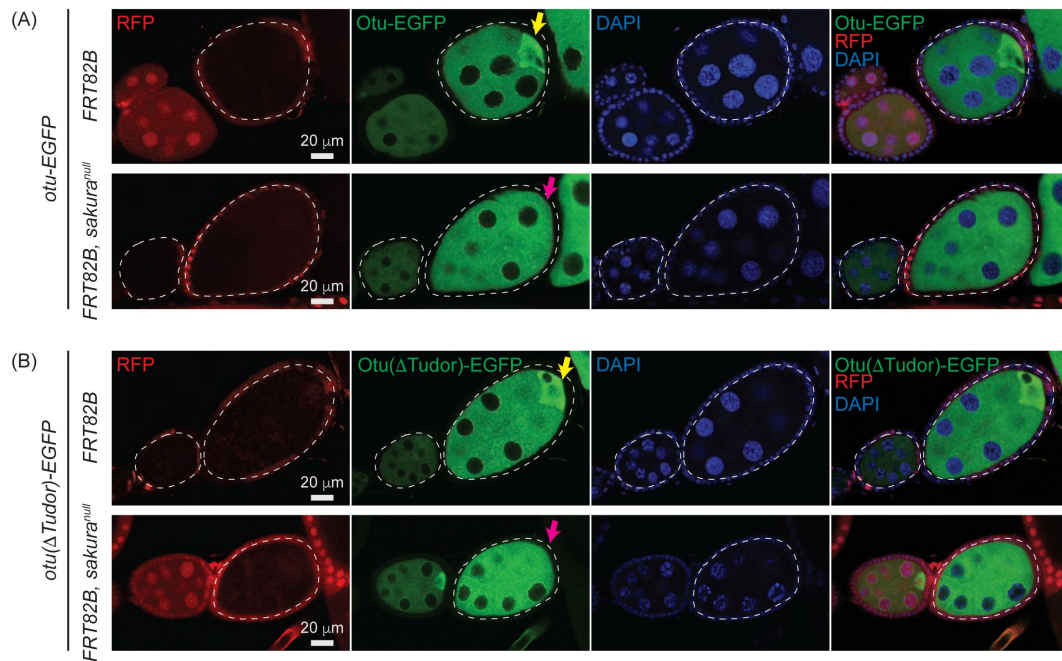


Fig 12.

Otu enrichment to the posterior within egg chambers is lost in *sakura*^{null}

Confocal images of egg chambers with germline clones of *sakura*^{null} expressing (A) Otu-EGFP or (B) Otu(ΔTudor)-EGFP. Fly genotypes used: *hs-flp/w; otu-EGFP/+; FRT82B, ubi-RFP/FRT82B*. *hs-flp/w; otu-EGFP/+; FRT82B, ubi-RFP/FRT82B, sakura*^{null}. *hs-flp/w; otu(i1Tudor)-EGFP/+; FRT82B, ubi-RFP/FRT82B*. *hs-flp/w; otu(i1Tudor)-EGFP/+; FRT82B, ubi-RFP/FRT82B, sakura*^{null}. RFP (red), Otu-EGFP or Otu(ΔTudor)-EGFP (green), and DAPI (blue). Scale bars: 20 μm. Marked clones (RFP-negative) are outlined with white dotted lines. Yellow arrows indicate normal posterior enrichment of Otu-EGFP and Otu(ΔTudor)-EGFP signal; magenta arrows indicate the loss of this posterior enrichment in *sakura*^{null} clones.

to tumor formation, but are ultimately *sakura*^{null}GSCs and GSC-like cells are lost through a cell death mechanism, as evidenced by the elevated levels of cleaved Caspase-3 in *sakura*^{null} ovaries (**Fig 3F**).

Dpp/BMP signaling governs GSC self-renewal and cystoblast differentiation by repressing *bam* in GSCs and de-repressing it in daughter cystoblasts (**Fig S1**). This process is mediated by the transcription factor Mad, which, when phosphorylated (pMad), translocates into the nucleus to repress *bam* transcription (Kirilly and Xie 2007; Kahney et al. 2019; Hinnant et al. 2020). Our findings indicate that Bam expression is not limited to 8-cell cysts but continues throughout the germarium in ovaries lacking *sakura* function (**Fig 7A** and **7B**). The misexpression of Bam in GSCs likely results from reduced levels of pMad in the GSCs (**Fig 7C**, and **7D**). Given the low pMad levels observed upon loss of *sakura*, we propose that *bam* misregulation in GSCs occurs primarily at the transcriptional level, although Bam is also known to be subject to post-transcriptional regulation (Pek et al. 2009). The continued expression of Bam throughout the germarium, well beyond the GSC niche, suggests a failure to terminate *bam* expression at the 16-cell cyst stage—a developmental transition when *bam* is normally turned off. Thus, *sakura* mutants likely exhibit two distinct molecular defects contributing *bam* overexpression: (1) impaired Dpp/BMP signaling within the GSC niche, and (2) a failure to downregulate *bam* at 16-cell stage. Given the limited space within the GSC niche, the latter defect may be the predominant contributor to the observed *bam* overexpression. The molecular mechanism that silences *bam* at the 16-cell cyst stage remains poorly understood, and how *sakura* loss perturbs this regulation is unclear. The *sakura* mutant may therefore serve as a valuable model to investigate the mechanism that normally suppresses *bam* expression at the 16-cell stage.

Transposons are mobile genetic elements that, if not silenced, can generate DNA damage and genomic instability (Levin and Moran 2011). piRNAs derived from transposons and other repeats can target and silence transposon RNAs to preserve genome integrity in germ cells (Siomi et al. 2011). Loss of piRNAs leads to transposon derepression, resulting in increased DNA damage, which subsequently triggers cell death (Kang et al. 2018; Moon et al. 2018). Genetic damage in germ cells can cause developmental defects and diseases that may be inherited by the next generation. Thus, elimination of defective germ cells is crucial for maintaining germline integrity of a species (Chu et al. 2014; Ota and Kobayashi 2020). We found that loss of *sakura* results in reduced piRNA levels and loss of piRNA-mediated transposon silencing in the germline (**Fig 4**). The observed apoptosis, indicated by elevated cleaved Caspase-3 levels in *sakura* mutant ovaries (**Fig 3F**), suggest that desilencing of transposon due to reduced piRNA levels likely results in increased DNA damage, triggering cell death. We speculate that the germless phenotype in *sakura* mutants may partly arise from an apoptotic germline elimination program activated to maintain germline integrity.

In addition to its expression in GSCs and cysts in the germarium, Sakura is also expressed in germline cells in later-stage egg chambers (**Fig 1E-F**) and is required at this stage for proper oogenesis. When Sakura is depleted from region 2b of the germarium onward—leaving GSCs and early cyst cells intact—females have significantly fewer stage 14 oocytes and lay significantly fewer eggs (**Fig 5**). The failure to produce stage 14 mature oocytes likely stems from cytoskeletal disorganization that disrupt oocyte development, as evidenced by mislocalization of Orb (**Fig 5E**, **S7**, and **S12**). Notably, piRNA pathway mutants also exhibit similar Orb mislocalization (Ohtani et al. 2013). Therefore, the reduced piRNA levels (**Fig 4**) may contribute to the oocyte development defects, including Orb mislocalization, in *sakura* mutants. (**Fig 5E**).

Sakura does not possess any known protein domains. To infer its function, we identified Otu, a protein known to be crucial for oogenesis, as a protein partner of Sakura. Mutations in *otu* gene lead to a range of ovarian phenotypes, including germ cell loss, tumorous egg chambers filled with undifferentiated germ cells, defects in germline sexual identity determination, abnormalities in nurse cell chromosome structure, and defects in oocyte determination (King and Riley 1982;

Storto and King 1988 [\[1\]](#); Pauli et al. 1993 [\[2\]](#); Glenn and Searles 2001 [\[3\]](#)). We showed that germline depletion of Otu via RNAi phenocopies loss of *sakura*. Similar to Sakura, loss of *otu* inhibits Dpp/BMP signaling, resulting in low pMad levels in GSCs and Bam de-repression (**Fig 11A-B** [\[4\]](#)). Additionally, loss of *otu* also results in the loss of piRNA-mediated silencing, paralleling the effects of *sakura* loss. Furthermore, germline knockdown of *otu* yielded a similar ratio of germless ovaries as seen in *sakura-RNAi*, and simultaneous knockdown of both *otu* and *sakura* exacerbated the germless ovary phenotype (**Fig S10** [\[5\]](#)). These observations raise the possibility that Sakura and Otu function together to regulate germ cell maintenance in the ovaries and are involved in Dpp/BMP signaling to balance stem cell renewal and differentiation.

Otu possesses deubiquitinase activity, catalyzed by its N-terminal Otu domain (Ji et al. 2017 [\[6\]](#); Ji et al. 2019 [\[7\]](#)). In germ cells, Otu interacts with Bam, forming a deubiquitinase complex that deubiquitinates and thereby stabilizes CycA, promoting GSC differentiation. The predicted structure of Sakura and Otu complex suggests that the Mid domain of Sakura directly contacts the Otu domain of Otu (**Fig 9A** [\[8\]](#)). We found that Sakura lacks a deubiquitinase activity and it does not directly affect Otu's deubiquitinase rate in our in vitro assays using Ub-Rhodamine 110 as a substrate (**Fig S16** [\[9\]](#)). This does not preclude the possibility that Sakura may still influence Otu's deubiquitinase activity, potentially guiding Otu to its substrate and determining substrate specificity. Our in vitro assays may not have detected such specificity changes.

Otu is an RNA-binding protein whose deubiquitinase activity is enhanced by RNA binding (Ji et al. 2019 [\[7\]](#)). Bam, along with other proteins such as Bgcn, Mei-P26, and Sxl, binds —a key stem cell maintenance factor (Wang and Lin 2004 [\[10\]](#))— and represses its translation in GSCs (Wang and Lin 2004 [\[10\]](#); Li et al. 2009 [\[11\]](#); Chau et al. 2012 [\[12\]](#); Li et al. 2013 [\[13\]](#)). Bam and Otu form a protein complex (Ji et al. 2017 [\[6\]](#)). Future studies should explore whether Sakura modulates Otu's RNA-binding properties and its interaction with other proteins. Although Sakura is exclusively expressed in ovaries, particularly in germline cells including GSCs (**Fig 1** [\[14\]](#)), Otu is broadly expressed in various tissues, including testes and gut (Steinhauer and Kalfayan 1992b [\[15\]](#)). This restricted expression pattern suggests that Sakura may serve as a female germline-specific cofactor that enhances or modifies Otu's molecular functions. Identifying Otu's RNA targets and deubiquitinase substrates beyond CycA, as well as determining how Sakura binding influences these activities, will be key to elucidating their roles in oogenesis. One intriguing possibility is that Sakura regulates Otu's activity toward piRNA pathway components, thereby contributing proper piRNA biogenesis and function. Additionally, the Sakura-Otu complex may directly regulate essential post-transcriptional processes such as *sxl* alternative splicing and translational control of other oogenic RNAs.

In summary, this study identifies and characterizes the previously unknown gene *sakura*, which is specifically expressed in female germ cells and is essential for oogenesis and female fertility. Together with its protein partner Otu, Sakura likely regulates germline cell fate, maintenance, and differentiation.

Materials and methods

Fly strains

We generated the *sakura*^{null} strain by introducing indels within the Sakura coding region using the CRISPR/Cas9 genome editing system, as we previously reported (Zhu et al. 2018a [\[16\]](#); Zhu et al. 2018b [\[17\]](#); Zhu et al. 2019b [\[18\]](#); Zhu and Fukunaga 2021 [\[19\]](#)). The transgenic *sakura-EGFP*, *otu-EGFP*, and *otu(ΔTudor)-EGFP* strains were created following previously published methods (Fukunaga et al. 2012 [\[20\]](#); Kandasamy and Fukunaga 2016 [\[21\]](#); Zhu and Fukunaga 2021 [\[19\]](#)). A fragment containing Sakura cDNA flanked by ~1 kbp upstream and ~1 kbp downstream genomic sequences was cloned.

Fragments containing Otu and Otu(Δ Tudor) cDNAs, flanked by ~2.5 kbp upstream and ~1 kbp downstream genomic sequences, were also cloned. The EGFP gene was fused in-frame to the C-terminus of the Sakura and Otu coding sequences within these fragments, which were then inserted into a pattB plasmid vector. The *sakura-EGFP* plasmid was integrated into the 51C1 site within the fly genome using the BDSC:24482 fly strain and the PhiC31 system, while the *otu-EGFP* and *otu(Δ Tudor)-EGFP* plasmids were integrated into the 25C6 site of the genome using the attP40 fly strain and the PhiC31 system.

The *sakura RNAi* #1 (VDRC: v39727) and #2 (VDRC: v103660), *y-RNAi* (v106068), *TOsk-Gal4* (v 314033), Burdock sensor [*UAS-Dcr2*; *NGT-Gal4*; *nosGal4-VP16*, *nos>NLS_GF⁺_lacZ_vas-3'UTR_burdock-target*] (v 313217) strains were from Vienna *Drosophila* Resource Center. *w-RNAi* (BDSC: 35573), *otu-RNAi* (BDSC: 34065), *bam-RNAi* (BDSC: 33631), *cycA-RNAi* (BDSC: 29313), *UAS-Dcr-2*; *NGT-Gal4* (BDSC: 25751), *FRT82B/TM6C*, *Sb* (BDSC: 86313), *otu¹⁴* (BDSC: 6025), *UASp-Mad-GFP* (BDSC: 604592), and *UASp-*tkv.Q253D** (BDSC: 604934) were obtained from the Bloomington Stock Center. The *bam-GFP* reporter (DGRC: 118177) and *vasa-EGFP* knocked-in fly (DGRC: 118616) were from Kyoto *Drosophila* Stock Center.

Fertility assay

The female fertility assay was performed as previously described (Zhu et al. 2018b [↗](#); Liao et al. 2019 [↗](#); Zhu et al. 2019a [↗](#); Zhu and Fukunaga 2021 [↗](#)). Briefly, five test virgin females of each test genotype were mated with three wild-type (OregonR) males in a cage with a 6-cm grape juice agar plate supplemented with wet yeast paste. The agar plates were replaced daily, and the number of eggs laid on the third plate (on day 3) was recorded. After incubation at 25°C for an additional day, the number of hatched eggs was counted. At least three cages per genotype were tested.

For the male fertility assay, a single test male was mated with five wild-type (OregonR) virgin females in each vial, following previously published methods (Zhu et al. 2018b [↗](#); Zhu et al. 2019a [↗](#); Zhu and Fukunaga 2021 [↗](#)). After 3 days, the females were transferred to a new vial (vial 1). Every two days, they were transferred to a new vial until a total of four vials were obtained. After 2 days in the fourth vial, the females were removed, and the total number of progenies emerging from these four vials was counted. At least five males per genotype were tested.

Sakura and Otu antibodies

We expressed a recombinant full-length Sakura protein as an N-terminal 6xHis-tagged protein in *E. coli* using a modified pET vector and purified it using Ni-sepharose (GE Healthcare) and HiTrapQ HP (GE Healthcare) columns (Fukunaga and Doudna 2009 [↗](#)). Recombinant Otu fragments (145-405aa and 406-853aa) were similarly expressed as N-terminally 6xHis-MBP-fusion proteins in *E. coli* and purified using Ni-sepharose. These purified proteins were used as antigens to generate polyclonal anti-Sakura and anti-Otu sera in rabbits (Pocono Rabbit Farm & Laboratory, Inc.). The rabbit polyclonal anti-Sakura antibodies were affinity purified using His-MBP-Sakura recombinant protein and Affigel-15 (Bio-rad), following the manufacturer's instructions. Rabbit anti-Otu sera were first pre-cleared with purified His-MBP protein bound to Affigel-15, and the unbound fraction was affinity purified using purified His-MBP-Otu fragments (145-405aa and 406-853aa) and Affigel-15.

Immunostaining

Stereomicroscope images of dissected ovaries were taken using Leica M125 stereomicroscope. Ovaries from 2- to 5-day-old, yeast-fed females were hand-dissected in 1X PBS (137 mM NaCl, 2.7 mM KCl, 10 mM Na₂HPO₄, 1.8 mM KH₂PO₄, pH 7.4) at room temperature. The dissected ovaries were fixed in a fixative buffer (4% formaldehyde, 15 mM PIPES (pH 7.0), 80 mM KCl, 20 mM NaCl₂, 2 mM EDTA, and 0.5 mM EGTA), incubated for 30 minutes at room temperature with gentle rocking. After fixation, the ovaries were rinsed three times with PBX (0.1% Triton X-100 in 1X PBS)

and then incubated in a blocking buffer (2% donkey serum in 3% BSA [w/v], 0.02% NaN₃ [w/v] in PBX) for 1 hour at room temperature. Then, the ovaries were incubated with primary antibodies diluted in the blocking buffer overnight at 4°C. The following day, the ovaries were rinsed three times with PBX and incubated with Alexa Fluor-conjugated secondary antibodies for 2 hours. The ovaries were rinsed three times with PBX and then mounted in VECTASHIELD® PLUS antifade mounting medium with DAPI (H-2000, Vector lab). Confocal images were acquired on a Zeiss LSM700 confocal microscope at the Johns Hopkins University School of Medicine Microscope Facility.

The primary antibodies used for immunostaining were mouse anti-HTS (1B1) (DSHB, AB_528070, dilution: 1/100), mouse anti-Bam (DSHB, AB_10570327, 1/20), mouse anti-CycA (DSHB, AB_528188, 1/100), rat anti-Vasa (DSHB, AB_760351, 1/100), rat anti-DE Cadherin (DCAD2) (DSHB, AB_528120, 1/20), rabbit anti-pMad (Cell Signaling, Phospho-SMAD1/5 (Ser463/465) mAb #9516, 1/200), and rabbit anti-cleaved caspase-3 (Cell Signaling, Cleaved Caspase-3 (Asp175) #9661, 1/200). Secondary antibodies used were Alexa Fluor 488 Donkey anti-Mouse IgG (ThermoFisher, A21202, 1/100), Alexa Fluor 594 Donkey anti-Rat IgG (ThermoFisher, A21209, 1/100), Alexa Fluor 594 Donkey anti-Mouse IgG (ThermoFisher, A21203, 1/100), and Alexa Fluor 594 Donkey anti-Rabbit IgG ThermoFisher, A21207, 1/100). Rhodamine phalloidin (ThermoFisher, R415, 1/100) was used to stain F-Actin.

Germline clonal analysis

We generated *sakura*^{null} mutant clones using FLP/FRT-mediated recombination (Rubin and Huynh 2015 [\[1\]](#)). To induce *sakura*^{null} GSC clones, 3-day-old female flies of the genotype *hs-flp/w*; +; *FRT82B*, *ubi-GFP/FRT82B*, *sakura*^{null} were heat-shocked at 37°C for 1 hour, twice daily, with an 8-hour interval between heat shocks. Female flies of the genotype *hs-flp/w*; +; *FRT82B*, *ubi-GFP/FRT82B* were used as controls. Ovaries were dissected and stained 4, 7, and 14 days after clone induction. To induce primordial germ cell (PGC) clones, early third-instar larvae were subjected to the same heat shock regime, then allowed to develop to adulthood. Ovaries were dissected and analyzed at 3 to 4 days after eclosion.

To induce *sakura*^{null} mutant clones in the presence of *otu-EGFP* or *otu(ΔTudor)-EGFP* transgenes, 3-day-old female flies with the genotype *hs-flp/w*; *otu-EGFP*/+; *FRT82B*, *ubi-RFP/FRT82B*, *sakura*^{null} and *hs-flp/w*; *otu(i1Tudor)-EGFP*/+; *FRT82B*, *ubi-RFP/FRT82B*, *sakura*^{null} were heat-shocked under the same conditions. Control genotypes were *hs-flp/w*; *otu-EGFP*/+; *FRT82B*, *ubi-RFP/FRT82B* and *hs-flp/w*; *otu(i1Tudor)-EGFP*/+; *FRT82B*, *ubi-RFP/FRT82B* were used as controls. Flies were dissected 3-4 days after clone induction.

Western blot

Lysates of hand-dissected ovaries and tissues were prepared by homogenizing in RIPA buffer (50 mM Tris-HCl [pH 7.4], 150 mM NaCl, 1% [v/v] IGEPAL CA-630, 0.1% [w/v] sodium dodecyl sulfate (SDS), 0.5% [w/v] sodium deoxycholate, 1 mM ethylenediaminetetraacetic acid (EDTA), 5 mM dithiothreitol, and 0.5 mM phenylmethylsulfonyl fluoride (PMSF)) (Kandasamy et al. 2017 [\[2\]](#); Zhu et al. 2018b [\[3\]](#)). The homogenates were centrifuged at 21,000g at 4°C for 10 min, and the protein concentration of the supernatant was determined using the BCA protein assay kit (Pierce) as needed. Fifteen µg of total protein was loaded per lane for Western blot. The sources and dilutions of the primary antibodies were as below. Rabbit anti-Sakura (1/10000, generated in this study), rabbit anti-Otu (1/10000, generated in this study), rabbit anti-alpha-Tubulin [EP1332Y] (1/10000, Abcam, ab52866), mouse anti-alpha-Tubulin [12G10] (1/10000, DSHB, AB_1157911), mouse anti-FLAG (1/10000, Sigma, F1804), mouse anti-HA (1/10000, Sigma, H3663), and mouse anti-GFP [GF28R] (1/3000, Invitrogen, 14-6674-82). IRDye 800CW goat anti-mouse IgG, IRDye 800CW goat anti-rabbit IgG, IRDye 680RD goat anti-mouse, and IgG IRDye 680RD goat anti-rabbit were used as secondary antibodies. The membranes were scanned using the Li-Cor Odyssey CLx Imaging System.

Mass spectrometry

Immunoprecipitation of Sakura-EGFP protein was performed using the GFP-Trap Magnetic Agarose Kit (Proteintech, gtmak-20) on dissected ovaries from flies harboring *sakura-EGFP* transgene, with *w¹¹¹⁸* flies as controls. Ovaries were homogenized in 200 μ L ice-cold lysis buffer (10 mM Tris-HCl [pH 7.5], 150 mM NaCl, 0.5 mM EDTA, 0.05% [v/v] IGEPAL CA-630) containing 1 $\times$ protease inhibitor cocktail (100 $\times$ protease inhibitor cocktail contains 120 mg/ml 1 mM 4-(2-aminoethyl) benzene sulfonyl fluoride hydrochloride (AEBSF), 1 mg/ml aprotinin, 7 mg/ml bestatin, 1.8 mg/ml E-64, and 2.4 mg/ml leupeptin). After homogenization, the tubes were placed on ice for 30 minutes, and the homogenates were extensively pipetted every 10 minutes. The lysates were then centrifuged at 17,000 $\times$ g for 10 minutes at 4°C. The supernatants were transferred to pre-chilled tubes, and 300 μ L dilution buffer (10 mM Tris/Cl pH 7.5, 150 mM NaCl, 0.5 mM EDTA) supplemented with 1 $\times$ protease inhibitor cocktail were added. The diluted lysates were then added to the GFP-trap magnetic beads in 1.5 mL tubes and rotated for 1 hour at 4°C. After separating the beads with a magnetic tube rack, the beads were washed three times with 500 μ L wash buffer (10 mM Tris/Cl pH 7.5, 150 mM NaCl, 0.05 % [v/v] IGEPAL CA-630). Proteins were eluted with 40 μ L acidic elution buffer (200 mM glycine pH 2.5) followed by immediate neutralization with 5 μ L neutralization buffer (1 M Tris pH 10.4).

As a quality control before mass spectrometry, $\sim$ 5 μ L of the samples were mixed with an equal volume of 2 $\times$ SDS-PAGE loading buffer (80 mM Tris-HCl [pH 6.8], 2% [w/v] SDS, 10% [v/v] glycerol, 0.0006% [w/v] bromophenol blue, 2% [v/v] 2-mercaptoethanol), heated at 95 °C for 3 min, and run on 4–20% Mini-PROTEAN® TGX™ Precast Protein Gels (Biorad, #4561094). Silver staining was then performed by using Pierce™ Silver Stain Kit (ThermoFisher, 24612) to assess the quality of the immunoprecipitated protein samples. Mass spectrometry was conducted at the Mass Spectrometry Core at the Department of Biological Chemistry, Johns Hopkins School of Medicine, as previously described (Zhu and Fukunaga 2021 [\[1\]](#)).

Co-immunoprecipitation

For co-immunoprecipitation of endogenous Sakura protein, wild-type ovaries (*w¹¹¹⁸*) were homogenized in RIPA buffer, centrifuged at 21,000g at 4°C for 10 min, and the clear supernatant protein lysates were used for immunoprecipitation. Four μ g of rabbit anti-Sakura and normal rabbit IgG (Cell Signaling, #2729) were incubated with 50 μ L of Dynabeads Protein G (ThermoFisher, 10004D) for 20 min at room temperature. The beads were washed once with PBST (1 $\times$ PBS with 0.1% Tween-20). The ovary lysate supernatant was then incubated with the washed beads at room temperature for 30 min, followed by three washes with PBST. The proteins were eluted with 2 $\times$ SDS-PAGE loading buffer (80 mM Tris-HCl [pH 6.8], 2% [w/v] SDS, 10% [v/v] glycerol, 0.0006% [w/v] bromophenol blue, 2% [v/v] 2-mercaptoethanol) and heated at 70°C for 10 min. After bead separation using a magnetic tube rack, the eluted proteins in 2 $\times$ SDS-PAGE loading buffer were heated again at 95°C for 3 min.

Transient protein expression in S2 cells was performed using the pAc5.1/V5-HisB plasmid vector (Invitrogen). A total of 1 μ g plasmids were transfected using the Effectene transfection reagent (Qiagen, 301425). Three days after transfection, cells were harvested and lysed with ice-cold lysis buffer supplemented with 1 $\times$ protease inhibitor cocktail. The cell lysates were then centrifuged at 17,000 $\times$ g for 10 min at 4°C, and the clear supernatants were collected for immunoprecipitation. For anti-HA immunoprecipitation, supernatants were incubated with 25 μ L (0.25 mg) of Pierce™ Anti-HA Magnetic Beads (ThermoFisher, 88837) at room temperature for 30 min. The beads were then washed three times with TBST (1 $\times$ TBS [0.05 M Tris/HCl and 0.15 M NaCl, pH 7.6] with 0.05% Tween-20). For anti-FLAG immunoprecipitation, 2 μ g of mouse anti-FLAG (1/10000, Sigma, F1804) was incubated with 50 μ L of Dynabeads Protein G (ThermoFisher, 10004D) for 10 min at room temperature. The beads were washed once with PBST. The S2 cell lysate supernatant was incubated with the washed beads at room temperature for 15 min. The beads were washed three

times with PBST. In both anti-HA and anti-FLAG immunoprecipitations, proteins were eluted with 2× SDS-PAGE loading buffer. For anti-HA, the beads in 2x SDS-PAGE loading buffer were heated at 95°C for 7 min. For anti-FLAG, the beads were heated at 70°C for 10 min, then were separated using a magnetic tube rack. The eluted proteins in 2x SDS-PAGE loading buffer were heated again at 95°C for 3 min.

Small RNA and mRNA sequencing

Poly-A⁺ mRNA purification was performed as previously described (Fukunaga et al. 2012 [DOI](#)). Small RNA libraries and poly-A⁺ mRNA libraries were prepared, sequenced on Hiseq2500 (Illumina), and analyzed, as previously reported (Fukunaga et al. 2014 [DOI](#); Kandasamy et al. 2017 [DOI](#); Liao et al. 2018 [DOI](#); Zhu et al. 2018a [DOI](#); Zhu et al. 2018b [DOI](#); Liao et al. 2019 [DOI](#); Zhu et al. 2019a [DOI](#); Zhu et al. 2019b [DOI](#)). SRA accession number for these datasets is PRJNA1156618.

RT-PCR

Total RNAs from testes and ovaries were prepared using miRVana (Thermo Fisher Scientific). RNAs were treated with Turbo DNase (Thermo Fisher Scientific) to remove potential genomic DNA contamination. A total of 1 µg of RNA was reverse-transcribed into cDNA using SuperScriptTM VILOTM MasterMix (Thermo Fisher Scientific). To assess *sxl* alternative splicing, PCR was performed using GoTaq Green Master Mix (Promega) with the primers *sxl*-F (CTCACCTTCGATCGAGGGTGTA) and *sxl*-R (GATGGCAGAGAATGGGAC).

In vitro deubiquitination assay

C-terminally 6xHis-tagged recombinant Sakura protein was expressed in *E. coli* using a modified pET vector and was purified using Ni-sepharose. N-terminally 3xHA-HRV3Csite-3xFLAG-tagged Otu protein and a negative control, N-terminally 3xHA-HRV3Csite-3xFLAG-tagged firefly luciferase protein, were expressed in S2 cells using pAc5.1/V5-HisB vector. The proteins were immunoprecipitated using PierceTM Anti-HA Magnetic Beads (ThermoFisher, 88837) as described previously, with a more stringent washing step to effectively remove interacting proteins. The beads were washed six times with a high salt wash buffer (TBST with 800 mM NaCl). The proteins were eluted by cleaving off 3xHA tag with 25 nM GST-HRV3C protease in a cleavage buffer (25mM Tris-HCl pH 7.4, 150 mM NaCl, 5% glycerol, 2mM EDTA), incubated at 4°C with rotation for 6 hours. An equal volume of 100% glycerol was added to the eluted protein.

Purified proteins, including Otu, luciferase, and Sakura, were mixed with Ub-Rhodamine 110 (Ubiquitin-Proteasome Biotechnologies, M3020) in a total volume of 30 µL in reaction buffer (20 mM Tris-HCl, pH 7.5, 200 mM NaCl, 5 mM MgCl₂, 2 mM DTT). The mixture was added to a black 384-well low-volume plate. Fluorescence measurements were taken using SpectraMax i3x Multi-Mode Microplate Reader at 37°C, with excitation and emission wavelengths set at 485/20 and 530/20 nm respectively. The fluorescence intensity for each condition was averaged from triplicates and plotted as a function of time.

Supplementary figures

Fig S1.

Schematic illustration of *bam* expression regulation.

Cap cells secrete diffusible Decapentaplegic (Dpp), which is received by its receptor, a heterodimer of Thick vein (Tkv) and Punt (Put), in GSCs. The activated Dpp signaling eventually phosphorylates Mother-against-dpp (Mad). The phosphorylated Mad (pMad) represses the transcription of *bag-of-marbles* (*bam*). The repression of *bam* in GSCs is crucial for maintaining their stemness. Cystoblasts do not receive Dpp, and Bam expression is crucial for promoting cystoblast differentiation from GSCs.

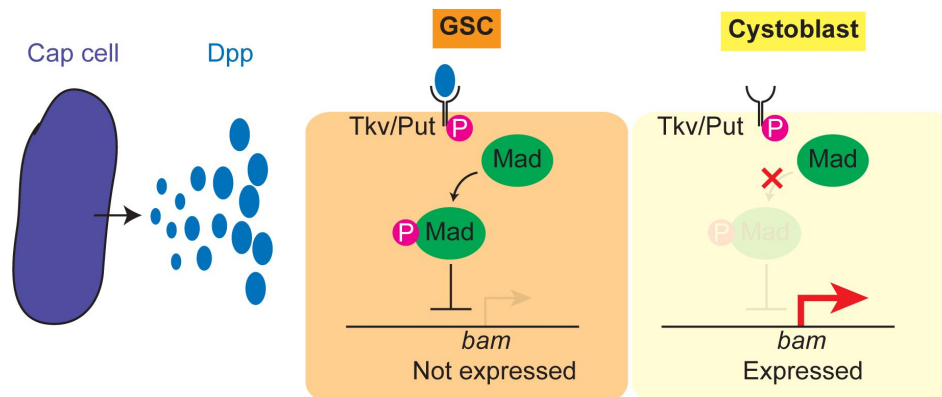


Fig S2.

***sakura* mRNA expression pattern.**

Data obtained from <http://flybase.org/reports/FBgn0040602.htm>.

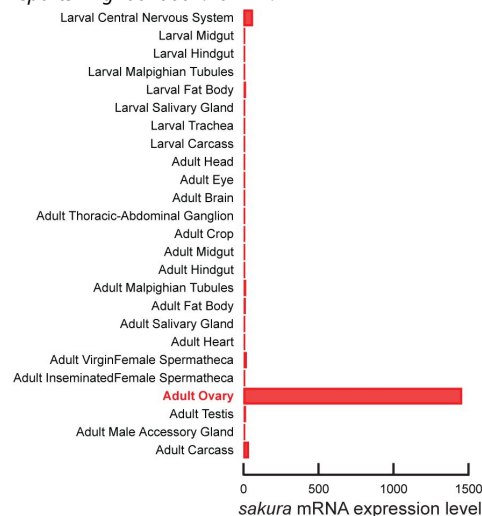


Fig S3.

Male fertility assay.

The numbers of the progeny flies obtained from crosses between test males and wild-type (OregonR) virgin females are shown. Mean $\pm$ SD (n = 5).

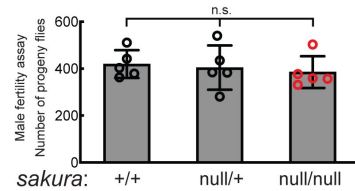
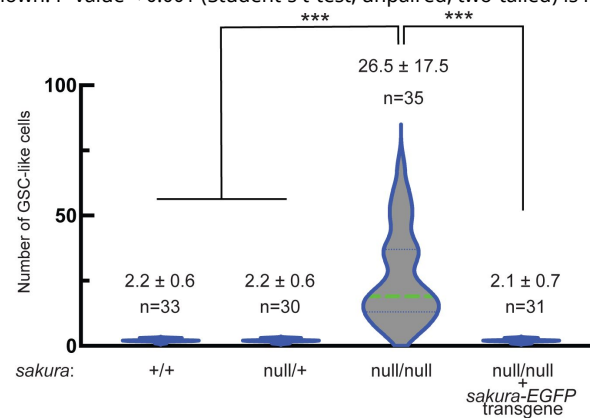


Fig S4.

sakura^{null} ovaries are tumorous

Violin plots of GSC-like cell numbers in germaria of indicated genotypes of 2-5 days old flies. Mean $\pm$ SD and the biological replicate number n are also shown. P-value < 0.001 (Student's t-test, unpaired, two-tailed) is indicated by ***.



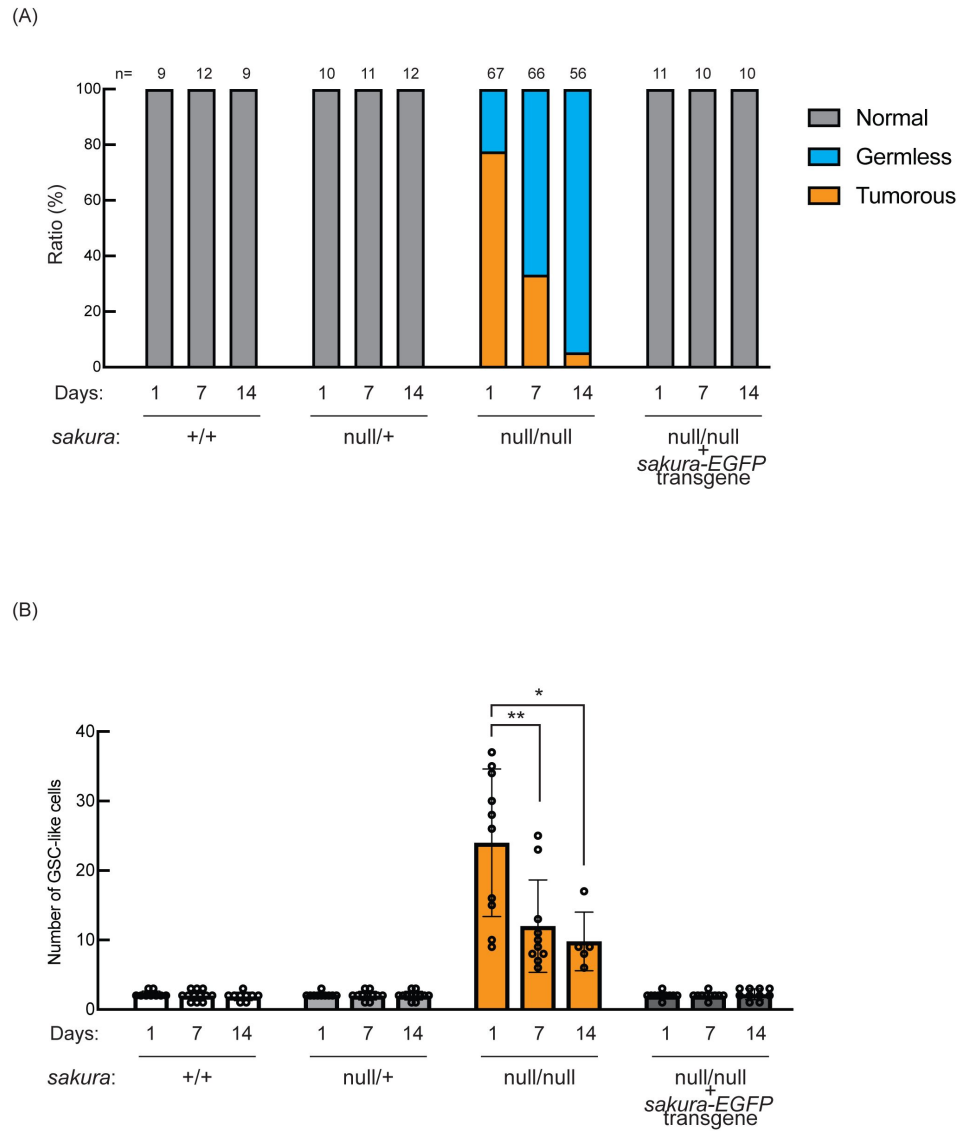


Fig S5.

The ratio of germless ovarioles increases over time in *sakura*^{null} ovaries

(A) Ratio (%) of normal, germless, and tumorous ovarioles of the indicated genotypes at 0-1 days, 7 days, and 14 days post-eclosion. (B) Quantification of GSC-like cell numbers in ovarioles of the indicated genotypes of at 0-1 days, 7 days, and 14 days post-eclosion. Germless ovarioles were excluded from this analysis.

Fig S6.

Sex-specific alternative splicing of *sxl* is dysregulated in *sakura*^{null} ovaries

Sex-specific alternative splicing of *sxl* was analyzed by RT-CPR followed by agarose gel electrophoresis and SYBR Safe staining. Ovaries from controls and *sakura-EGFP* rescue flies showed the expected female-specific *sxl* isoform, while testes from control flies exhibited the male-specific isoform. *sakura*^{null} ovaries showed increased expression of the male-specific isoform and reduced levels of female-specific isoform.

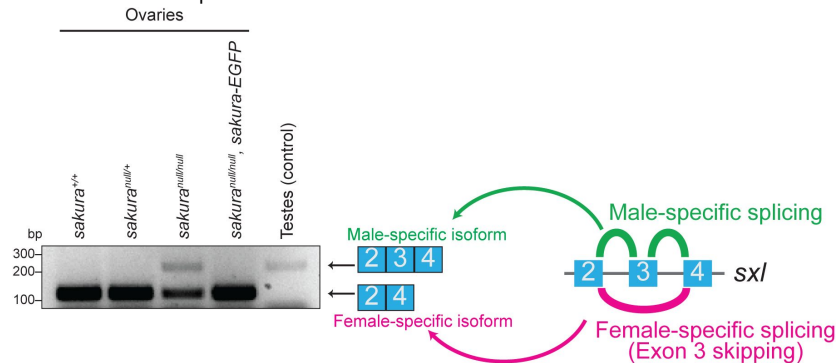


Fig S7.

sakura is important for oogenesis in later-stage egg chambers

Confocal images of ovaries from *sakura* RNAi knockdown flies (*UAS-Dcr2, TOSK-Gal4 > sakura*^{RNAi}) stained with phalloidin (F-Actin) and anti-Orb antibodies. F-Actin (red), Orb (green), and DAPI (blue). Yellow arrows indicate normal Orb enrichment at the posterior of developing oocytes. White arrowheads indicate egg chambers exhibiting cytoskeletal disorganization. Scale bars: 50 μ m.

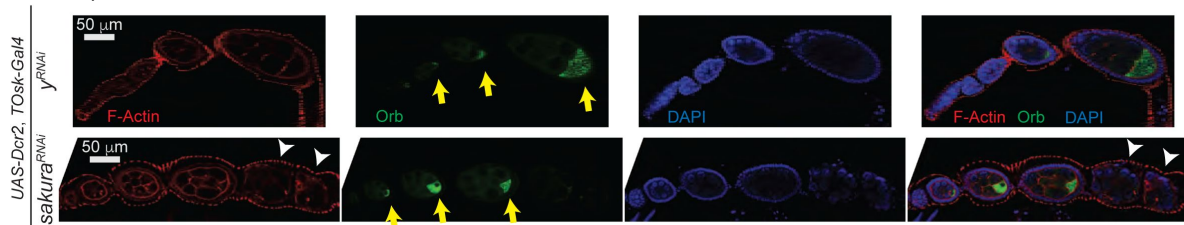


Fig S8.

sakura^{null} clone germline cells intrinsically cause tumorous phenotype

Number of GFP-positive GSC-like cells in germaria with marked (GFP-negative) GSCs of the indicated genotypes at 4, 7, and 14 days after clone induction. GSC-like cells containing round spectrosome were identified through immunostaining with anti-Hts antibody.

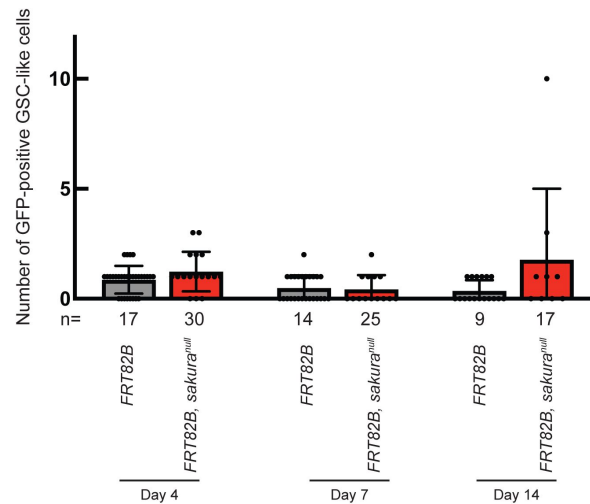
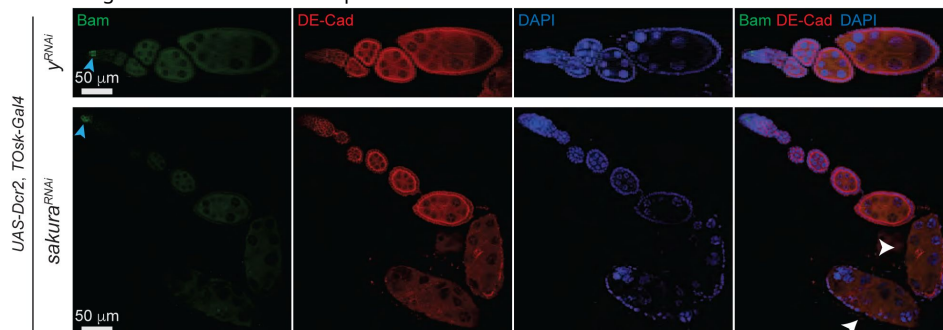


Fig S9.

Bam is not misexpressed in *TOsk-Gal4* > *sakura*^{RNAi} egg chambers.

Confocal images of ovaries from *sakura* RNAi knockdown flies driven by *UAS-Dcr2* and *TOsk-Gal4*, stained with anti-Bam and anti-DE-Cadherin (DE-Cad) antibodies. *y-RNAi* was used as a control. Bam (green), DE-Cad (red), and DAPI (blue). Cyan arrowheads indicate high Bam expression in 8-cell cysts within the germarium. White arrowheads indicate egg chambers with cytoskeletal disorganization. Scale bars: 50 μ m.



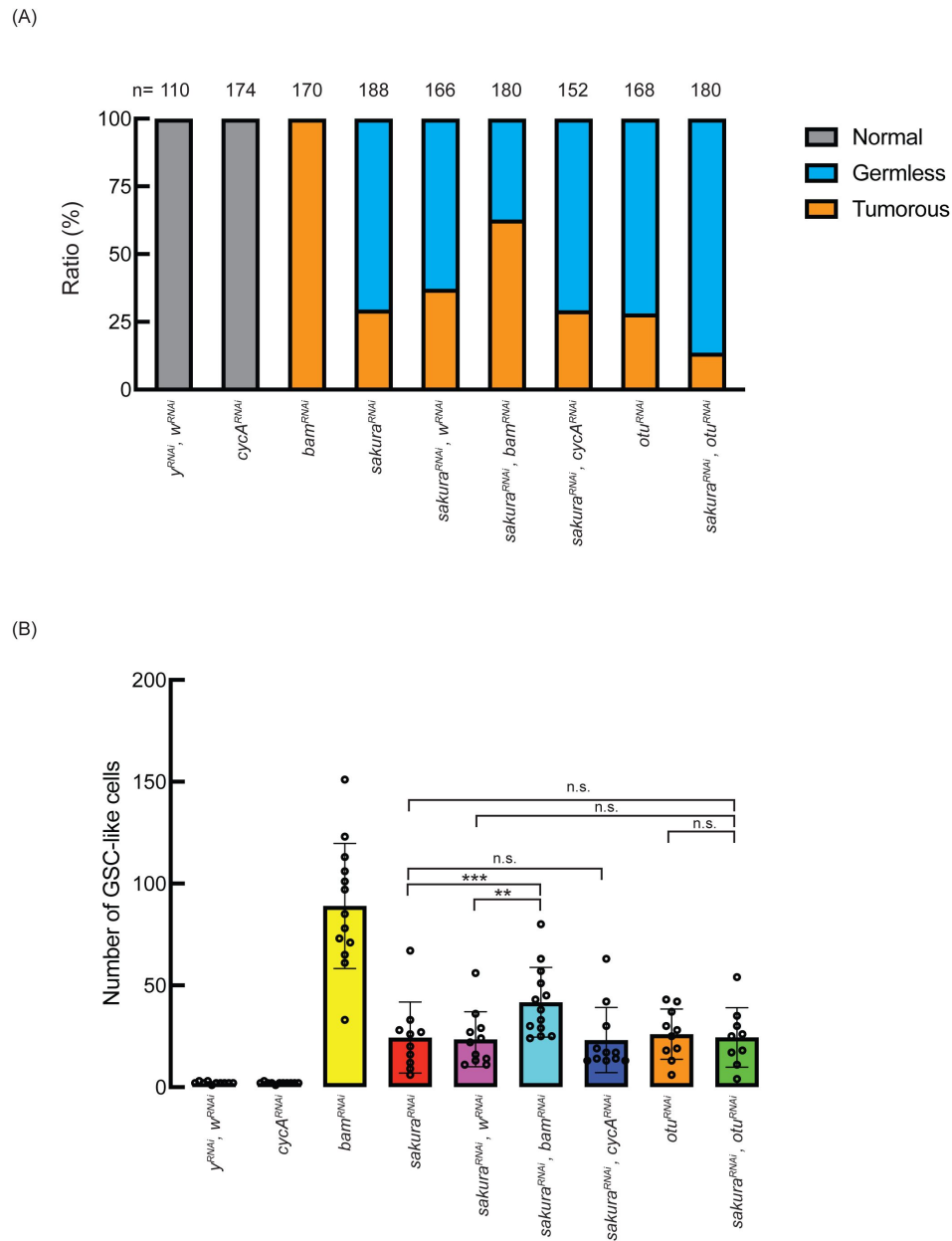


Fig S10.

Ratio of germless and tumorous phenotypes from double RNAi knockdown of *sakura* with *bam*, *cycA*, or *otu*

(A) Ratio (%) of normal, germless, and tumorous ovarioles in double RNAi knockdown of *sakura* with *bam*, *cycA*, or *otu*. *UAS-Dcr2* and *NGT-Gal4* was used to drive RNAi knockdown in the germline. n is the total number of ovarioles examined for the indicated genotypes of 2-5 day-old flies. (B) Quantification of GSC-like cell number in germaria of indicated genotypes of 2-5 day-old flies. Germless ovarioles were excluded from this analysis.

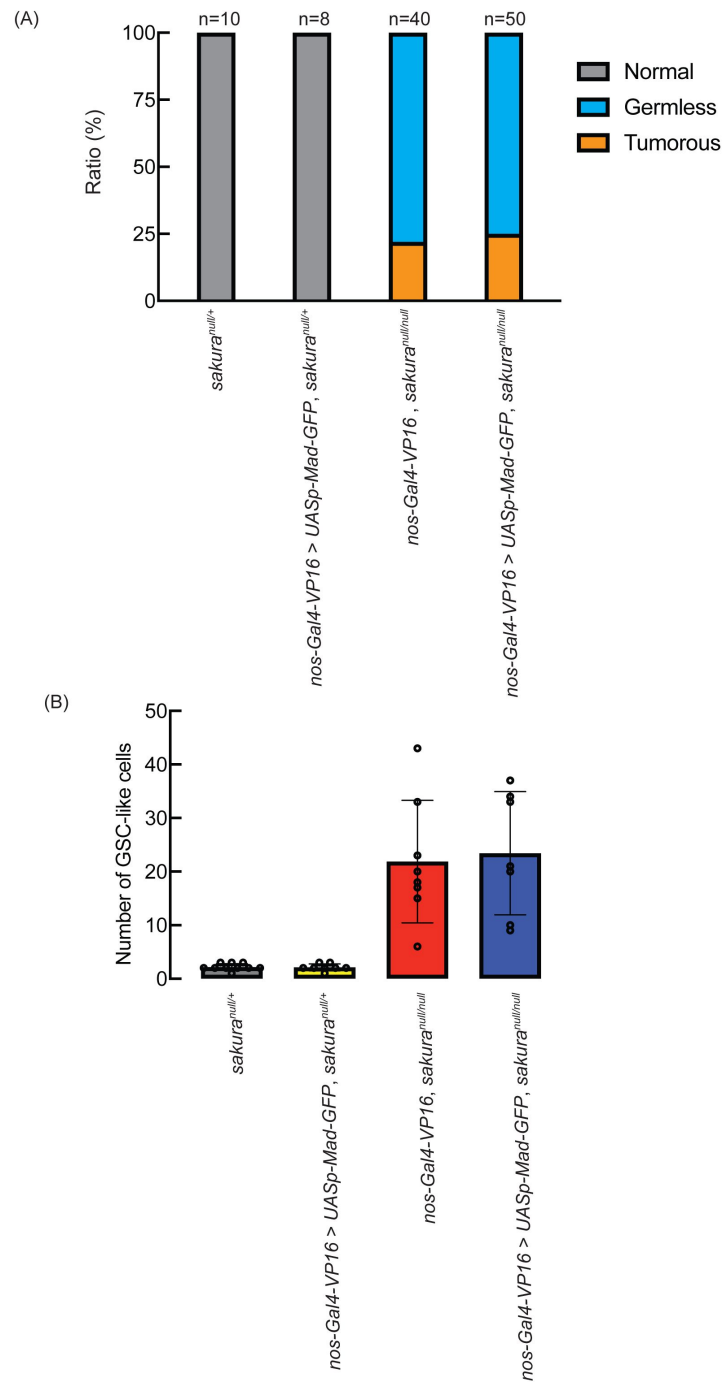


Fig S11.

***nos-Gal4-VP16 > UASp-Mad-GFP* did not rescue the *sakura^{null}* phenotypes**

(A) Ratio (%) of normal, germless, and tumorous ovarioles in the indicated genotypes of 2-5 day-old flies. n denotes the total number of ovarioles examined for each genotype. (B) Quantification of GSC-like cell number in germaria of the indicated genotypes from 2-5 day-old flies. Germless ovarioles were excluded from this analysis.

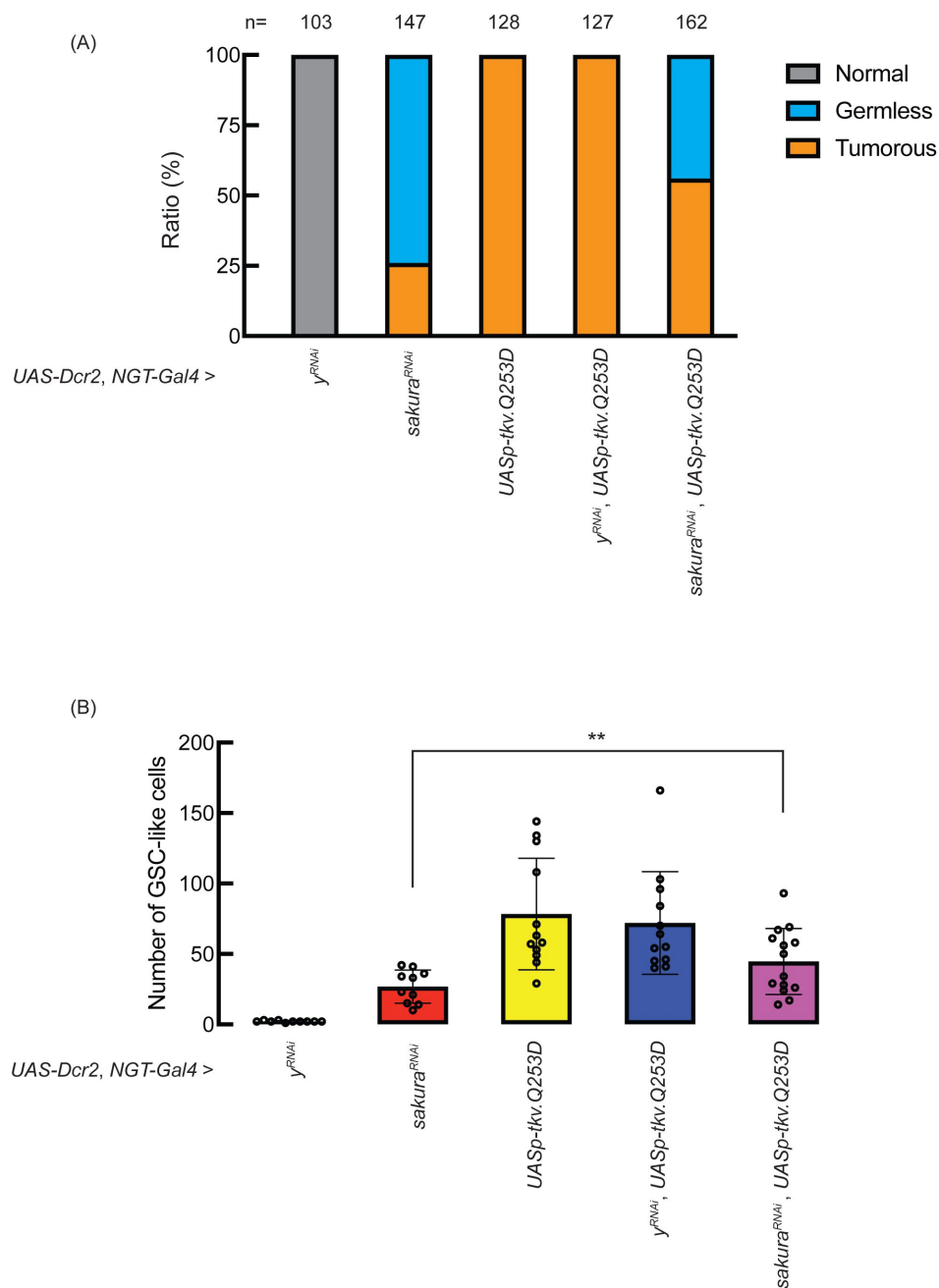


Fig S12.

Ratio of germless and tumorous phenotypes in *sakura* RNAi knockdown and *NGT-Gal4* > *UASp-tkv.Q253D*.

(A) Ratio (%) of normal, germless, and tumorous ovarioles in the indicated genotypes of 2-5 day-old flies. RNAi knockdown and UASp-tkv.Q253D expression were driven with *UAS-Dcr2* and *NGT-Gal4*. n indicates the total number of ovarioles examined for each genotype. (B) Quantification of GSC-like cell number in germaria of the indicated genotypes from 2-5 day-old flies. Germless ovarioles were excluded from this analysis.

Fig S13.

anti-Otu Western blot of dissected ovary lysates.

(A) Western blot of dissected ovary lysates. (B) Western blot of dissected ovary lysates prepared from flies 2-5 hours and 3-7 days post eclosion and co-IP with anti-Sakura. The SDS-PAGE gel was run for a longer duration to better separate the 104 kDa and 98 kDa Otu isoforms. In 2-5 hour ovaries, the 104 kDa Otu isoform is more abundant, while in 3-7 day ovaries, the 98 kDa isoform predominates. Both Otu isoforms co-IPed with Sakura. The same 3-7 day ovary input and IP samples used for **Fig 9B** were used for anti-Otu and anti-Alpha-Tubulin Western. The anti-Sakura image is the same as shown in **Fig 9B**.

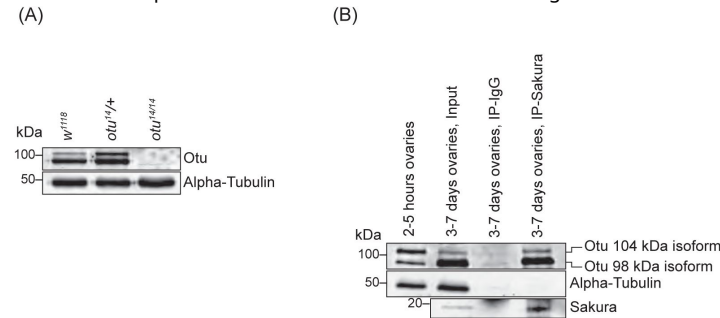


Fig S14.

Co-immunoprecipitations assay to test interaction between N-terminal fragments of Sakura and Otu in S2 cells

Co-immunoprecipitation assay using beads bound with anti-FLAG antibody followed by Western blotting. S2 cell lysates expressing mCherry-3xHA were used as controls. This is the reciprocal co-immunoprecipitation of **Fig 9F**.

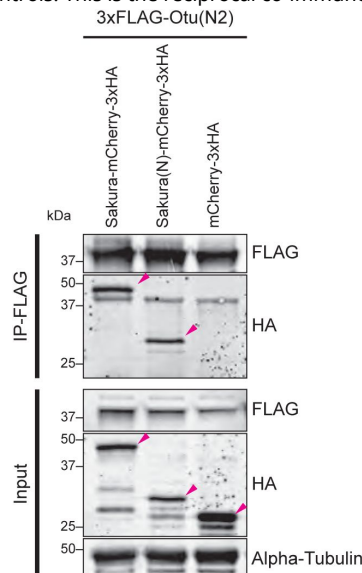


Fig S15.

Depletion of Sakura does not deplete Otu and depletion of Otu does not deplete Sakura

(A) Western blot of dissected ovary lysates. The same samples used for Fig 5B were used for anti-Otu Western. The anti-Sakura and anti-Alpha tubulin images are the same as shown in Fig 5B. (B) The same samples used for Fig 10C were used for anti-Sakura Western. The anti-Otu and anti-Alpha tubulin images are the same as shown in Fig 10C.

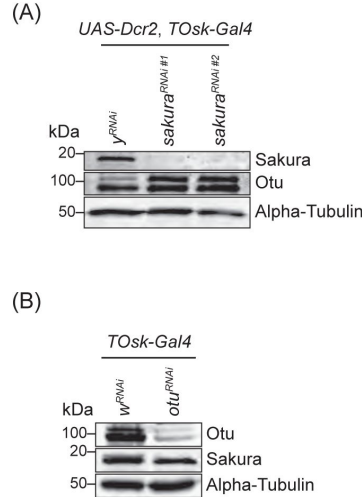
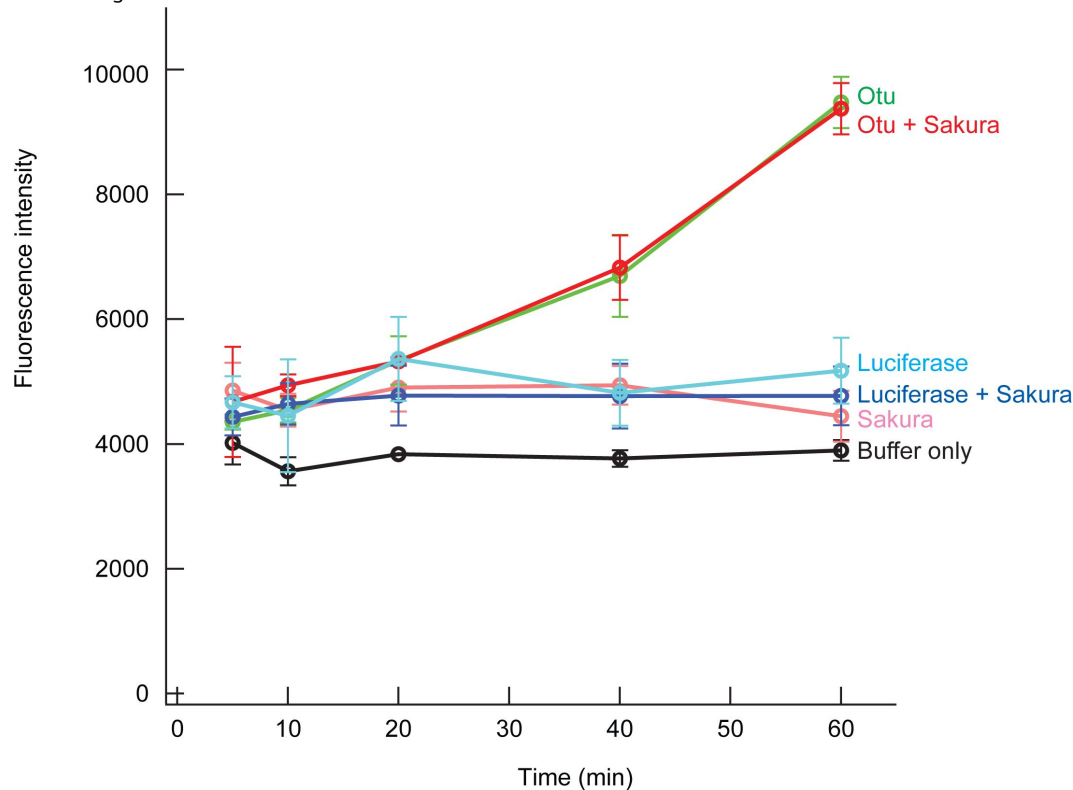


Fig S16.

In vitro deubiquitination assay.

The mean fluorescence intensity of three replicates was plotted. Error bars (+/-) are standard deviations. Firefly Luciferase was used as a negative control.



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Additional information

Author Contributions

Methodology and Investigation, A.A., L.Z., and R.F.; Conceptualization and Writing, A.A. and R.F.; Funding Acquisition and Supervision, R.F.

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Reviewer #1 (Public review):

Summary:

In this manuscript, Azlan et al. identified a novel maternal factor called Sakura that is required for proper oogenesis in *Drosophila*. They showed that Sakura is specifically expressed in the female germline cells. Consistent with its expression pattern, Sakura functioned autonomously in germline cells to ensure proper oogenesis. In sakura KO flies, germline cells were lost during early oogenesis and often became tumorous before degenerating by apoptosis. In these tumorous germ cells, piRNA production was defective and many transposons were derepressed. Interestingly, Smad signaling, a critical signaling pathway for the GSC maintenance, was abolished in sakura KO germline stem cells, resulting in ectopic expression of Bam in whole germline cells in the tumorous germline. A recent study reported that Bam acts together with the deubiquitinase Otu to stabilize Cyc A. In the absence of sakura, Cyc A was upregulated in tumorous germline cells in the germarium. Furthermore, the authors showed that Sakura co-immunoprecipitated Otu in ovarian extracts. A series of in vitro assays suggested that the Otu (1-339 aa) and Sakura (1-49 aa) are sufficient for their direct interaction. Finally, the authors demonstrated that the loss of otu phenocopies the loss of sakura, supporting their idea that Sakura plays a role in germ cell maintenance and differentiation through interaction with Otu during oogenesis.

Strengths:

To my knowledge, this is the first characterization of the role of CG14545 genes. Each experiment seems to be well-designed and adequately controlled

Weaknesses:

However, the conclusions from each experiment are somewhat separate, and the functional relationships between Sakura's functions are not well established. In other words, although the loss of Sakura in the germline causes pleiotropic effects, the cause-and-effect relationships between the individual defects remain unclear.

Comments on latest version:

The authors have attempted to address my initial concerns with additional experiments and refutations. Unfortunately, my concerns, especially my specific comments 1-3, remain unaddressed. The present manuscript is descriptive and fails to describe the molecular mechanism by which Sakura exerts its function in the germline. Nevertheless, this reviewer acknowledges that the observed defects in sakura mutant ovaries and the possible physiological significance of the Sakura-Out interaction are worth sharing with the research community, as they may lay the groundwork for future research in functional analysis.

<https://doi.org/10.7554/eLife.103828.2.sa3>

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Reviewer #3 (Public review):

In this very thorough study, the authors characterize the function of a novel *Drosophila* gene, which they name Sakura. They start with the observation that sakura expression is predicted to be highly enriched in the ovary and they generate an anti-sakura antibody, a line with a GFP-tagged sakura transgene, and a sakura null allele to investigate sakura localization and function directly. They confirm the prediction that it is primarily expressed in the ovary and, specifically, that it is expressed in germ cells, and find that about 2/3 of the mutants lack germ cells completely and the remaining have tumorous ovaries. Further investigation reveals that Sakura is required for piRNA-mediated repression of transposons in germ cells. They also

find evidence that sakura is important for germ cell specification during development and germline stem cell maintenance during adulthood. However, despite the role of sakura in maintaining germline stem cells, they find that sakura mutant germ cells also fail to differentiate properly such that mutant germline stem cell clones have an increased number of "GSC-like" cells. They attribute this phenotype to a failure in the repression of Bam by dpp signaling. Lastly, they demonstrate that sakura physically interacts with otu and that sakura and otu mutants have similar germ cell phenotypes. Overall, this study helps to advance the field by providing a characterization of a novel gene that is required for oogenesis. The data are generally high-quality and the new lines and reagents they generated will be useful for the field.

Comments on latest version:

With these revisions, the authors have addressed my main concerns.

<https://doi.org/10.7554/eLife.103828.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

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In this manuscript, Azlan et al. identified a novel maternal factor called Sakura that is required for proper oogenesis in Drosophila. They showed that Sakura is specifically expressed in the female germline cells. Consistent with its expression pattern, Sakura functioned autonomously in germline cells to ensure proper oogenesis. In Sakura KO flies, germline cells were lost during early oogenesis and often became tumorous before degenerating by apoptosis. In these tumorous germ cells, piRNA production was defective and many transposons were derepressed. Interestingly, Smad signaling, a critical signaling pathway for GSC maintenance, was abolished in sakura KO germline stem cells, resulting in ectopic expression of Bam in whole germline cells in the tumorous germline. A recent study reported that Bam acts together with the deubiquitinase Otu to stabilize Cyc A. In the absence of sakura, Cyc A was upregulated in tumorous germline cells in the germarium. Furthermore, the authors showed that Sakura co-immunoprecipitated Otu in ovarian extracts. A series of in vitro assays suggested that the Otu (1-339 aa) and Sakura (1-49 aa) are sufficient for their direct interaction. Finally, the authors demonstrated that the loss of otu phenocopies the loss of sakura, supporting their idea that Sakura plays a role in germ cell maintenance and differentiation through interaction with Otu during oogenesis.

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However, the conclusions from each experiment are somewhat separate, and the functional relationships between Sakura's functions are not well established. In other words, although the loss of Sakura in the germline causes pleiotropic effects, the cause-and-effect relationships between the individual defects remain unclear.

Reviewer #2 (Public review):

In this study, the authors identified CG14545 (and named it Sakura), as a key gene essential for Drosophila oogenesis. Genetic analyses revealed that Sakura is vital for both oogenesis progression and ultimate female fertility, playing a central role in the renewal and differentiation of germ stem cells (GSC).

The absence of Sakura disrupts the Dpp/BMP signaling pathway, resulting in abnormal bam gene expression, which impairs GSC differentiation and leads to GSC loss. Additionally, Sakura is critical for maintaining normal levels of piRNAs. Also, the authors convincingly demonstrate that Sakura physically interacts with Otu, identifying the specific domains necessary for this interaction, suggesting a cooperative role in germline regulation. Importantly, the loss of otu produces similar defects to those observed in Sakura mutants, highlighting their functional collaboration.

The authors provide compelling evidence that Sakura is a critical regulator of germ cell fate, maintenance, and differentiation in Drosophila. This regulatory role is mediated through the modulation of pMad and Bam expression. However, the phenotypes observed in the germarium appear to stem from reduced pMad levels, which subsequently trigger premature and ectopic expression of Bam. This aberrant Bam expression could lead to increased CycA levels and altered transcriptional regulation, impacting piRNA expression. Given Sakura's role in pMad expression, it would be insightful to investigate whether overexpression of Mad or pMad could mitigate these phenotypic defects (UAS-Mad line is available at Bloomington Drosophila Stock Center).

As suggested reviewer 1, we tested whether overexpression of Mad could rescue or mitigate the loss of sakura phenotypic defects, by using *nos-Gal4-VP16 > UASp-Mad-GFP* in the background of *sakura^{null}*. As shown in Fig S11, we did not observe any mitigation of defects.

Then, we also tested whether expressing a constitutive active form of Tkv, by using *UAS-Dcr2, NGT-Gal4 > UASp-tkv.Q235D* in the background of *sakura^{RNAi}*. As shown in Fig S12, we did not observe any mitigation of defects by this approach either.

A major concern is the overstated role of Sakura in regulating Orb. The data does not reveal mislocalized Orb; rather, a mislocalized oocyte and cytoskeletal breakdown, which may be secondary consequences of defects in oocyte polarity and structure rather than direct misregulation of Orb. The conclusion that Sakura is necessary for Orb localization is not supported by the data. Orb still localizes to the oocyte until about stage 6. In the later stage, it looks like the cytoskeleton is broken down and the oocyte is not positioned properly, however, there is still Orb localization in the ~8-stage egg chamber in the oocyte. This phenotype points towards a defect in the transport of Orb and possibly all other factors that need to localize to the oocyte due to cytoskeletal breakdown, not Orb regulation directly. While this result is very interesting it needs further evaluation on the underlying mechanism. For example, the decrease in E-cadherin levels leads to a similar phenotype and Bam is known to regulate E-cadherin expression. Is Bam expressed in these later knockdowns?

We examined Bam and DE-Cadherin expression in later RNAi knockdowns driven by *ToskGal4*. As shown in Fig S9, Bam was not expressed in these later knockdowns compared with controls. DE-Cadherin staining suggested a disorganized structure in late-stage egg chambers.

We agree that we overstated a role of Sakura in regulating Orb in the initial manuscript. We changed the text to avoid overstating.

The manuscript would benefit from a more balanced interpretation of the data concerning Sakura's role in Orb regulation. Furthermore, a more expanded discussion on Sakura's potential role in pMad regulation is needed. For example, since Otu and Bam are involved in translational regulation, do the authors think that Mad is not translated and therefore it is the reason for less pMad? Currently the discussion presents just a summary of the results and not an extension of possible interpretation discussed in context of present literature.

We changed the text to avoid overstating a role of Sakura in regulating Orb localization.

Based on our newly added results showing that transgenic overexpression of Mad could not rescue or mitigate the phenotypic defects of *sakura*^{null} mutant (Fig S11), we do not think the reason for less pMad is less translation of Mad.

Reviewer #3 (Public review):

In this very thorough study, the authors characterize the function of a novel Drosophila gene, which they name Sakura. They start with the observation that sakura expression is predicted to be highly enriched in the ovary and they generate an anti-sakura antibody, a line with a GFP-tagged sakura transgene, and a sakura null allele to investigate sakura localization and function directly. They confirm the prediction that it is primarily expressed in the ovary and, specifically, that it is expressed in germ cells, and find that about 2/3 of the mutants lack germ cells completely and the remaining have tumorous ovaries. Further investigation reveals that Sakura is required for piRNA-mediated repression of transposons in germ cells. They also find evidence that sakura is important for germ cell specification during development and germline stem cell maintenance during adulthood. However, despite the role of sakura in maintaining germline stem cells, they find that sakura mutant germ cells also fail to differentiate properly such that mutant germline stem cell clones have an increased number of "GSC-like" cells. They attribute this phenotype to a failure in the repression of Bam by dpp signaling. Lastly, they demonstrate that sakura physically interacts with otu and that sakura and otu mutants have similar germ cell phenotypes. Overall, this study helps to advance the field by providing a characterization of a novel gene that is required for oogenesis. The data are generally high-quality and the new lines and reagents they generated will be useful for the field. However, there are some weaknesses and I would recommend that they address the comments in the Recommendations for the authors section below.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

General Comments:

(1) The gene nomenclature: As mentioned in the text, Sakura means cherry blossom and is one of the national flowers of Japan. I am not sure whether the phenotype of the CG14545 mutant is related to Sakura or not. I would like to suggest the authors reconsider the naming.

The striking phenotype of *sakura* mutant is tumorous and germless ovarioles. The tumorous phenotype, exhibiting lots of round fusome in germarium visualized by anti-Hts staining, looks like cherry blossom blooming to us. Also, the germless phenotype reminds us falling of the cherry blossom, especially considering that the ratio of tumorous phenotype decreases and that of germless decreases over fly age. Furthermore, "Sakura" symbolizes birth and renewal in Japanese culture (the last author of this manuscript is Japanese). Our findings indicated that the gene *sakura* is involved in regulation of renewal and differentiation of

GSCs (which leads to birth). These are the reasons for the naming, which we would like to keep.

(2) In many of the microscopic photographs in the figures, especially for the merged confocal images, the resolution looks low, and the images appear blurred, making it difficult to judge the authors' claims. Also, the Alpha Fold structure in Figure 10A requires higher contrast images. The magnification of the images is often inadequate (e.g. Figures 3A, 3B, 5E, 7A, etc). The authors should take high-magnification images separately for the germarium and several different stages of the egg chambers and lay out the figures.

We are very sorry for the low-resolution images. This was caused when the original PDF file with high-resolution images was compressed in order to meet the small file size limit in the eLife submission portal. In the revised submission, we used high-resolution images.

Specific Comments

(1) How Sakura can cooperate with Otu remains unanswered. Sakura does not regulate deubiquitinase activity in vitro. Both sakura and otu appear to be involved in the Dpp-Smad signaling pathway and in the spatial control of Bam expression in the germarium, whereas Otu has been reported to act in concert with Bam to deubiquitinate and stabilize Cyc A for proper cystoblast differentiation. Therefore, it is plausible that the stabilization of Cyc A in the Sakura mutant is an indirect consequence of Bam misexpression and independent of the Sakura-Otu interaction. The authors may need to provide much deeper insight into the mechanism by which Sakura plays roles in these seemingly separable steps to orchestrate germ cell maintenance and differentiation during early oogenesis.

Yes, it is possible that the stabilization of CycA in the *sakura* mutant is an indirect consequence of Bam misexpression and independent of the Sakura-Otu interaction. To test the significance and role of the Sakura-Otu interaction, we have attempted to identify Sakura point mutants that lose interaction with Otu. If such point mutants were successfully obtained, we were planning to test if their transgene expression could rescue the phenotypes of *sakura* mutant as the wild-type transgene did. However, after designing and testing the interaction of over 30 point mutants with Otu, we could not obtain such mutant version of Sakura yet. We will continue making efforts, but it is beyond the scope of the current study. We hope to address this important point in future studies.

(2) Figure 3A and Figure 4: The authors show that piRNA production is abolished in Sakura KO ovaries. It is known that piRNA amplification (the ping-pong cycle) occurs in the Vasa-positive perinuclear nuage in nurse cells. Is the nuage normally formed in the absence of Sakura? The authors provide high-magnification images in the germarium expressing Vas-GFP. How does Sakura, and possibly Otu, contribute to piRNA production? Are the defects a direct or indirect consequence of the loss of Sakura?

We provided higher magnification images of germarium expressing Vasa-EGFP in *sakura* mutant background (Fig 3A and 3B). The nuage formation does not seem to be dysregulated in *sakura* mutant. Currently, we do not know if the piRNA defects are direct or indirect consequence of the loss of Sakura. This question cannot be answered easily. We hope to address this in future studies.

(3) Figure 7 and Figure 12: The authors showed that Dpp-Smad signaling was abolished in Sakura KO germline cells. The same defects were also observed in otu mutant ovaries (Figure 12B). How does the Sakura-Otu axis contribute to the Dpp-Smad pathway in the germline?

As we mentioned in the response to comment (1), we attempted to test the significance and role of the Sakura-Otu interaction, including in the Dpp-Smad pathway in the germline, but we have not yet been able to obtain loss-of-interaction mutant(s) of Sakura. We hope to address this in future studies.

(4) Figure 9 and Fig 10: The authors raised antibodies against both Sakura and Otu, but their specificities were not provided. For Western blot data, the authors should provide whole gel images as source data files. Also, the authors argue that the Otu band they observed corresponds to the 98-kDa isoform (lines 302-304). The molecular weight on the Western blot alone would be insufficient to support this argument.

When we submitted the initial manuscript, we also submitted original, uncropped, and unmodified whole Western blot images for all gel images to the eLife journal, as requested. We did the same for this revised submission. I believe eLife makes all those files available for downloading to readers.

In the newly added Fig S13B, we used very young 2-5 hours ovaries and 3-7 days ovaries. 2-5 days ovaries contain only mostly pre-differentiated germ cells. Older ovaries (3-7 days in our case here) contain all 14 stages of oogenesis and later stages predominate in whole ovary lysates.

As reported in previous literature (Sass et al. 1995), we detected a higher abundance of the 104 kDa Otu isoform than the 98 kDa isoform in from 2-5 hours ovaries and predominantly the 98 kDa isoform in 3-7 days ovaries (Fig S13B). These results confirmed that the major Otu isoform we detected in Western blot, all of which uses old ovaries except for the 2-5 hours ovaries in Fig S13B, is the 98 kDa isoform.

(5) Otu has been reported to regulate ovo and Sxl in the female germline. Is Sakura involved in their regulation?

We examined *sxl* alternative splicing pattern in *sakura* mutant ovaries. As shown in Fig S6, we detected the male-specific isoform of *sxl* RNA and a reduced level of the female-specific *sxl* isoform in *sakura* mutant ovaries. Thus Sakura seems to be involved in *sxl* splicing in the female germline, while further studies will be needed to understand whether Sakura has a direct or indirect role here.

(6) Lines 443-447: The GSC loss phenotype in piwi mutant ovaries is thought to occur in a somatic cell-autonomous manner: both piwi-mutant germline clones and germline-specific piwi knockdown do not show the GSC-loss phenotype. In contrast, the authors provide compelling evidence that Sakura functions in the germline. Therefore, the Piwi-mediated GSC maintenance pathway is likely to be independent of the Sakura-Otu axis.

We changed the text accordingly.

Reviewer #2 (Recommendations for the authors):

Overall, this is a cleanly written manuscript, with some sentences/sections that are confusing the way they are constructed (i.e. Line 37-38, 334, section on Flp/FRT experiments).

We rewrote those sections to avoid confusion.

Comment for all merged image data: the quality of the merged images is very poor - the individual channels are better but should also be reprocessed for more resolved image

data sets. Also, it would be helpful to have boundaries drawn in an individual panel to identify the regions of the germarium, as cartooned in Figure S1A (which should be brought into Figure 1) F-actin or Vsg staining would have helped throughout the manuscript to enhance the visualization of described phenotypes.

We are very sorry for the low-resolution images. This was caused when the original PDF file with high-resolution images was compressed in order to meet the small file size limit in the eLife submission portal. In the revised submission, we used high-resolution images.

We outlined the germarium in Fig 1E.

We brought the former FigS1 into Fig 1A.

We provided Phalloidin (F-Actin) staining images in Fig S7.

All p-values seem off. I recommend running the data through the student t-test again.

We used the student t-test to calculate p-values and confirmed that they are correct. We don't understand why the reviewer thinks all p-values seem off.

In the original manuscript, as we mentioned in each figure legends, we used asterisk (*) to indicate p-value <0.05, without distinguishing whether it's <0.001, <0.01 or <0.05.

Probably reviewer 2 is suggesting us to use ***, **, and *, to indicate p-value of <0.001, <0.01, and <0.05, respectively? If so, we now followed reviewer2's suggestions.

Figure 1

(1) Within the text, C is mentioned before A.

We updated the text and now we mentioned Fig 1A before Fig 1C.

(2) B should be the supplemental figure.

We moved the former Fig 1B to Supplemental Figure 1.

(3) C - How were the different egg chamber stages selected in the WB? Naming them 'oocytes' is deceiving. Recommend labeling them as 'egg chambers', since an oocyte is claimed to be just the one-cell of that cyst.

We changed the labeling to egg chambers.

(4) Is the antibody not detecting Sakura in IF? There is no mention of this anywhere in the manuscript.

While our Sakura antibody detects Sakura in IF, it seems to detect some other proteins as well. Since we have Sakura-EGFP fly strain (which fully rescues *sakura^{null}* phenotypes) to examine Sakura expression and localization without such non-specific signal issues, we relied on Sakura-EGFP rather than anti-Sakura antibodies for IF.

(5) Expand on the reliance of the sakura-EGFP fly line. Does this overexpression cause any phenotypes?

sakura-EGFP does not cause any phenotypes in the background of sakura[+/+] and sakura[+/-].

(6) Line 95 "as shown below" is not clear that it's referencing panel D.

We now referenced Fig 1D.

(7) Re: Figures 1 E and F. There is no mention of Hts or Vasa proteins in the text. "Sakura-EGFP was not expressed in somatic cells such as terminal filament, cap cells, escort cells, or follicle cells (Figure 1E). In the egg chamber, Sakura-EGFP was detected in the cytoplasm of nurse cells and was enriched in developing oocytes (Figure 1F)". Outline these areas or label these structures/sites in the images. The color of Merge labels is confusing as the blue is not easily seen.

We mentioned Hts and Vasa in the text. We labeled the structures/sites in the images and updated the color labeling.

Figure 2

(1) Entire figure is not essential to be a main figure, but rather supplemental.

We don't agree with the reviewer. We think that the female fertility assay data, where *sakura* null mutant exhibits strikingly strong phenotype, which was completely rescued by our Sakura-EGFP transgene, is very important data and we would like to present them in a main figure.

(2) 2A- one star (*) significance does not seem correct for the presented values between 0 and 100+.

In the original manuscript, as we mentioned in each figure legends, we used asterisk (*) to indicate p-value <0.05, without distinguishing whether it's <0.001, <0.01 or <0.05.

Probably reviewer 2 is suggesting us to use ***, **, and *, to indicate p-value of <0.001, <0.01, and <0.05, respectively? If so, we now followed reviewer2's suggestions.

(3) 2C images are extremely low quality. Should be presented as bigger panels.

We are very sorry for the low-resolution images. This was caused when the original PDF file with high-resolution images was compressed in order to meet the small file size limit in the eLife submission portal. In the revised submission, we used high-resolution images. We also presented as bigger panels.

Figure 3

(1) "We observed that some *sakura*^{null} /null ovarioles were devoid of germ cells ("germless"), while others retained germ cells (Fig 3A)" What is described is, that it is hard to see. Must have a zoomed-in panel.

We provided zoomed-in panels in Fig 3B

(2) C - The control doesn't seem to match. Must zoom in.

We provided matched control and also zoomed in.

(3) For clarity, separate the tumorous and germless images.

In the new image, only one tumorous and one germless ovarioles are shown with clear labeling and outline, for clarity.

(4) Use arrows to help clearly indicate the changes that occur. As they are presented, they are difficult to see.

We updated all the panels to enhance clarity.

(5) Line 158 seems like a strong statement since it could be indirect.

We softened the statement.

Figure 4

(1) Line 188-189 - Conclusion is an overstatement.

We softened the statement.

(2) Is the piRNA reduction due to a change in transcription? Or a direct effect by Sakura?

We do not know the answers to these questions. We hope to address these in future studies.

Figure 5

(1) D - It might make more sense if this graph showed % instead of the numbers.

We did not understand the reviewer's point. We think using numbers, not %, makes more sense.

(2) Line 213 - explain why RNAi 2 was chosen when RNAi 1 looks stronger.

Fly stock of RNAi line 2 is much healthier than RNAi line 1 (without being driven Gal4) for some reasons. We had a concern that the RNAi line 1 might contain an unwanted genetic background. We chose to use the RNAi 2 line to avoid such an issue.

(3) In Line 218 there's an extra parenthesis after the PGC acronym.

We corrected the error.

(4) TOsk-Gal4 fly is not in the Methods section.

We mentioned TOsk-Gal4 in the Methods.

Figure 6:

(1) The FLP-FRT section must be rewritten.

We rewrote the FLP-FRT section.

(2) A - include statistics.

We included statistics using the chi-square test.

(3) B - is not recalled in the Results text.

We referred Fig 6B in the text.

| (4) Line 232 references Figure 3, but not a specific panel.

We referred Fig 3A, 3C, 3D, and 3E, in the text.

| Figure 7/8 - can go to Supplemental.

We moved Fig 8 to supplemental. However, we think Fig 7 data is important and therefore we would like to present them as a main figure.

| (1) There should be CycA expression in the control during the first 4 divisions.

Yes, there is CycA expression observed in the control during the first 4 divisions, while it's much weaker than in *sakura*^{null} clone.

| (2) Helpful to add the dotted lines to delineate (A) as well.

We added a dotted outline for germarium in Fig 7A.

| (3) Line 263 CycA is miswritten as CyA.

We corrected the typo.

| Figure 9

| (1) Otu antibody control?

We validated Otu antibody in newly added Fig 10C and Fig S13A.

| (2) Which Sakura-EGFP line was used? *sakura* het. or null background? This isn't mentioned in the text, nor legend.

We used Sakura-EGFP in the background of *sakura*[+/+]. We added this information in the methods and figure legend.

| (3) C - Why the switch to S2 cells? Not able to use the Otu antibody in the IP of ovaries?

We can use the Otu antibody in the IP of ovaries. However, in anti-Sakura Western after anti-Otu IP, antibody light chain bands of the Otu antibodies overlap with the Sakura band. Therefore, we switched to S2 cells to avoid this issue by using an epitope tag.

| Figure 10

| (1) A- The resolution of images of the ribbon protein structure is poor.

We are very sorry for the low-resolution images. This was caused when the original PDF file with high-resolution images was compressed in order to meet the small file size limit in the eLife submission portal. In the revised submission, we used high-resolution images.

| (2) A table summarizing the interactions between domains would help bring clarity to the data presented.

We added a table summarizing the fragment interaction results.

| (3) Some images would be nice here to show that the truncations no longer colocalize.

We did not understand the reviewer's points. In our study, even for the full-length proteins.

We have not shown any colocalization of Sakura and Otu in S2 cells or in ovaries, except that they both are enriched in developing oocytes in egg chambers.

Figure 12

(1) A - control and RNAi lines do not match.

We provided matched images.

(2) In general, since for Sakura, only its binding to Otu was identified and since they phenocopy each other, doesn't most of the characterization of Sakura just look at Otu phenotypes? Does Sakura knockdown affect Otu localization or expression level (and vice versa)?

We tested this by Western (Fig S15) and IF (Fig 12). Sakura knockdown did not decrease Otu protein level, and Otu knockdown did not decrease Sakura protein level (Fig S15). In *sakura^{null}* clone, Otu level was not notably affected (Fig 12). In *sakura^{null}* clone, Otu lost its localization to the posterior position within egg chambers.

Figure S6

(1) It is Luciferase, not Lucifase.

We corrected the typo.

Reviewer #3 (Recommendations for the authors):

(1) It is interesting that germless and tumorous phenotypes coexist in the same population of flies. Additional consideration of these essentially opposite phenotypes would significantly strengthen the study. For example, do they co-exist within the same fly and are the tumorous ovarioles present in newly eclosed flies or do they develop with age? The data in Figure 8 show that bam knockdown partially suppresses the germless phenotype. What effect does it have on the tumorous phenotype? Is transposon expression involved in either phenotype? Do Sakura mutant germline stem cell clones overgrow relative to wild-type cells in the same ovariole? Does sakura RNAi driven by NGT-Gal4 only cause germless ovaries or does it also cause tumorous phenotypes? What happens if the knockdown of Sakura is restricted to adulthood with a Gal80ts? It may not be necessary to answer all of these questions, but more insight into how these two phenotypes can be caused by loss of sakura would be helpful.

We performed new experiments to answer these questions.

do they co-exist within the same fly and are the tumorous ovarioles present in newly eclosed flies or do they develop with age?

Tumorous and germless ovarioles coexist in the same fly (in the same ovary). Tumorous ovarioles are present in very young (0-1 day old) flies, including newly eclosed (Fig S5). The ratio of germless ovarioles increases and that of tumorous ovarioles decreases with age (Fig S5).

The data in Figure 8 show that bam knockdown partially suppresses the germless phenotype. What effect does it have on the tumorous phenotype?

bam knockdown effect on tumorous phenotype is shown in Fig S10. *bam* knockdown increased the ratio of tumorous ovarioles and the number of GSC-like cells.

Is transposon expression involved in either phenotype?

Since our transposon-piRNA reporter uses germline-specific *nos* promoter, it is expressed only in germ line cells, so we cannot examine in germless ovarioles.

Do Sakura mutant germline stem cell clones overgrow relative to wild-type cells in the same ovariole?

Yes, Sakura mutant GSC clones overgrow. Please compare Fig 6C and Fig S8.

Does sakura RNAi driven by NGT-Gal4 only cause germless ovaries or does it also cause tumorous phenotypes?

Fig S10 and Fig S12 show the ovariole phenotypes of *sakura* RNAi driven by NGT-Gal4. It causes both germless and tumorous phenotypes.

What happens if the knockdown of Sakura is restricted to adulthood with a Gal80ts?

Our mosaic clone was induced at the adult stage, so we already have data of adulthood-specific loss of function. Gal80ts does not work well with *nos*-Gal4.

(2) The idea that the excessive *bam* expression in tumorous ovaries is due to a failure of *bam* repression by *dpp* signaling is not well-supported by the data. *Dpp* signaling is activated in a very narrow region immediately adjacent to the niche but the images in Figure 7A show *bam* expression in cells that are very far away from the niche. Thus, it seems more likely to be due to a failure to turn *bam* expression off at the 16-cell stage than to a failure to keep it off in the niche region. To determine whether *bam* repression in the niche region is impaired, it would be important to examine cells adjacent to the niche directly at a higher magnification than is shown in Figure 7A.

We provided higher magnification images of cells adjacent to the niche in new Fig 7A.

We found that cells adjacent to the niche also express Bam-GFP.

That said, we agree with the reviewer. A failure to turn *bam* expression off at the 16-cell stage may be an additional or even a main cause of *bam* misexpression in *sakura* mutant. We added this in the Discussion.

(3) In addition, several minor comments should be addressed:

a. Does anti-Sakura work for immunofluorescence?

While our Sakura antibody detects Sakura in IF, it seems to detect some other proteins as well. Since we have Sakura-EGFP fly strain to examine Sakura expression and localization without such non-specific signal issues, we relied on Sakura-EGFP rather than anti-Sakura antibodies.

b. Please provide insets to show the phenotypes indicated by the different color stars in Figure 3C more clearly.

We provided new, higher-magnification images to show the phenotypes more clearly.

c. Please indicate the frequency of the expression patterns shown in Figure 4D (do all ovarioles in each genotype show those patterns or is there variable penetrance?).

We indicated the frequency.

d. An image showing T_{Os}kGal4 driving a fluorophore should be provided so that readers can see which cells express Gal4 with this driver combination.

It has been already done in the paper ElMaghraby et al, GENETICS, 2022, 220(1), iyab179, so we did not repeat the same experiment.

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